English (1001CJA101021230067)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
19-05-2024

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $\mathbf{g = 10\ m/s^2}$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

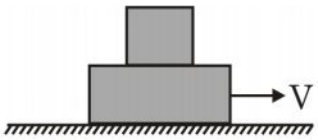
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

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1. A block is resting on a horizontal plate in the XY plane and co-efficient of friction between block and the plate is μ. The plate begins to move with velocity $v = bt^2$ in X-direction. At what time will the block starts sliding from plate ?



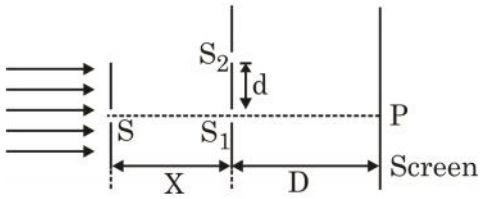
(A) $\frac{\mu b}{g}$

(B) $\frac{\mu bg}{2}$

(C) $\frac{\mu g}{b}$

(D) $\frac{\mu g}{2b}$

2. Monochromatic light of wavelength λ emerging from slit S illuminates slits S_1 and S_2 which are placed with respect to S as shown in figure. The distances x and D are large compared to the separation d between the slits. If x = D/2, the minimum value of d so that there is a dark fringe at the centre P of the screen is :-



(A) $\sqrt{\frac{\lambda D}{3}}$

(B) $\sqrt{\frac{2\lambda D}{3}}$

(C) $\sqrt{\lambda D}$

(D) $2\sqrt{\frac{\lambda D}{3}}$

3. An electromagnetic wave of frequency f = 3.0 MHz passes from vacuum into a dielectric medium with permittivity e = 4.0. Then :-

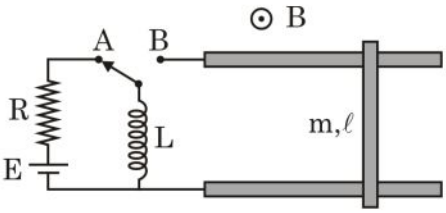
(A) wavelength is doubled and the frequency remains unchanged

(B) wavelength is doubled and frequency becomes half.

(C) wavelength is halved and frequency remains unchanged.

(D) wavelength and frequency both remain unchanged.

4. Switch is in position A for a long time. It is suddenly switched to the side B. Mass of rod is m and length is ℓ. Maximum speed of rod will be :-



(A) $\left(\frac{E}{R}\right)\sqrt{\frac{L}{m}}$

(B) $\left(\frac{E}{R}\right)\sqrt{\frac{m}{L}}$

(C) $2\left(\frac{E}{R}\right)\sqrt{\frac{L}{m}}$

(D) $2\left(\frac{E}{R}\right)\sqrt{\frac{m}{L}}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. The current density in a wire of radius a varies with radial distance r as $J = kr^2$, where k is a constant.

(A) Total current passing through the cross section of the wire is $I = \frac{\pi k a^4}{2}$.

(B) Total current passing through the cross section of the wire is $I = \frac{3\pi k a^3}{2}$

(C) The field at a distance $r > a$ is $B = \frac{\mu_0 k a^4}{4r}$

(D) The field at a distance $r < a$ is $B = \frac{\mu_0 k r^3}{4}$

6. When the number of electrons striking the target material in Coolidge's tube (i.e., X-ray tube) is increased keeping the potential difference same, then :-

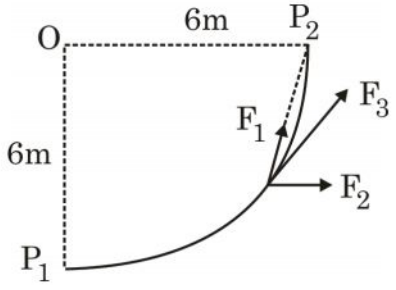
(A) intensity of X-rays increase

(B) wavelength of X-rays photons increase

(C) frequency of cut off X-rays photons remains same.

(D) energy of most energetic X-rays photons increase.

7. A smooth track in the form of a quarter circle of radius 6m lies in the vertical plane. A particle moves from P_1 to P_2 under the action of forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$. Force $\vec{F}_1$ is always toward P_2 and always 20 N in magnitude. Force $\vec{F}_2$ is always acts horizontally and is always 30 N in magnitude. Force $\vec{F}_3$ always acts tangentially to the track and is of magnitude 15 N. Select the correct alternative (s) :



(A) Work done by $\vec{F}_1$ is 120 J

(B) Work done by $\vec{F}_2$ is 180 J

(C) Work done by $\vec{F}_3$ is 45 π J

(D) $\vec{F}_1$ is conservative in nature

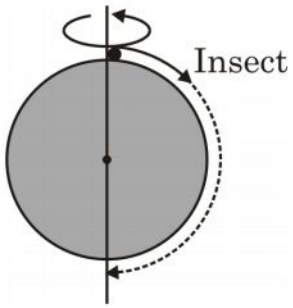
SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases.

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8. A solid sphere is rotating about an axis as shown in Figure. An insect follows the dotted path on the circumference of sphere as shown. Match the following :-



	List–I		List–II
(P)	Moment of inertia	(1)	will remain constant
(Q)	Angular velocity	(2)	will first increase then decrease
(R)	Angular momentum	(3)	will first decrease then increases
(S)	Rotational kinetic energy	(4)	will continuously decrease
		(5)	will continuously increase

(A) P $\longrightarrow$ 2;Q $\longrightarrow$ 3;R $\longrightarrow$ 1;S $\longrightarrow$ 3

(B) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 3;S $\longrightarrow$ 4

(C) P $\longrightarrow$ 5;Q $\longrightarrow$ 2;R $\longrightarrow$ 3;S $\longrightarrow$ 1

(D) P $\longrightarrow$ 3;Q $\longrightarrow$ 2;R $\longrightarrow$ 1;S $\longrightarrow$ 4

9. Match the following

	List–I		List–II
(P)	$\frac{R}{L}$	(1)	Time
(Q)	CR	(2)	Frequency
(R)	$\frac{E}{B}$	(3)	Speed
(S)	$\frac{1}{\mu_0 \varepsilon_0}$	(4)	Kinetic energy
		(5)	Kinetic energy per unit mass

(A) P ⟶ 3;Q ⟶ 2;R ⟶ 5;S ⟶ 1

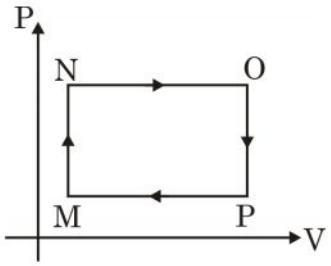
(B) P ⟶ 2;Q ⟶ 1;R ⟶ 3;S ⟶ 5

(C) P ⟶ 1;Q ⟶ 2;R ⟶ 3;S ⟶ 4

(D) P ⟶ 5;Q ⟶ 3;R ⟶ 1;S ⟶ 4

10. With reference to the following P-V diagram, match the following quantities.

[W : work done by gas ; ΔU : change in internal energy of gas]



	List–I		List–II
(P)	Process MN	(1)	W = 0
(Q)	Process NO	(2)	W < 0
(R)	Process OP	(3)	W > 0
(S)	Process PM	(4)	$\Delta u > 0$
		(5)	$\Delta u < 0$

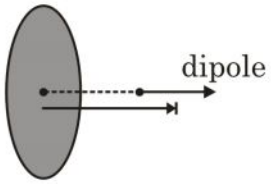
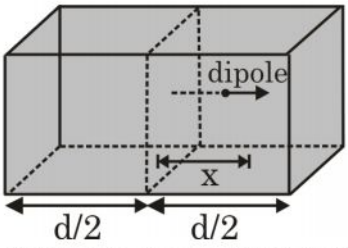
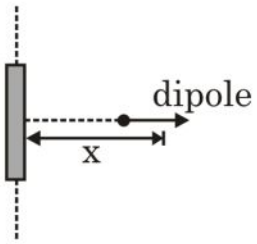
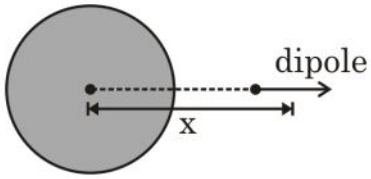
(A) P ⟶ 1,3;Q ⟶ 2,4;R ⟶ 1,5;S ⟶ 1,4

(B) P ⟶ 1,4;Q ⟶ 3,4;R ⟶ 2,3;S ⟶ 3,4

(C) P ⟶ 1,4;Q ⟶ 3,4;R ⟶ 1,5;S ⟶ 2,5

(D) P ⟶ 1,2;Q ⟶ 3,4;R ⟶ 1,5;S ⟶ 2,5

11. List–I shows different charge distributions and short electric dipole at a distance x from the charge distributions. List–II gives the dependence of force acting on the dipole as of function of x.

	List–I		**List–II**
(P)	 Uniformly charged thin disc.	(1)	x^{-2}
(Q)	 Uniformly charged thick infinite sheet. Dipole is placed inside the sheet at distance x from axis of sheet.	(2)	x^{-5}
(R)	 Uniform infinite line of charge.	(3)	x^{0}
(S)	 Uniformly charged sphere. Dipole is placed outside the sphere.	(4)	None of these
		(5)	x^{-3}

(A) P ⟶ 4;Q ⟶ 1;R ⟶ 3;S ⟶ 2

(B) P ⟶ 4;Q ⟶ 3;R ⟶ 1;S ⟶ 5

(C) P ⟶ 1;Q ⟶ 2;R ⟶ 4;S ⟶ 3

(D) P ⟶ 4;Q ⟶ 3;R ⟶ 1;S ⟶ 2

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases.

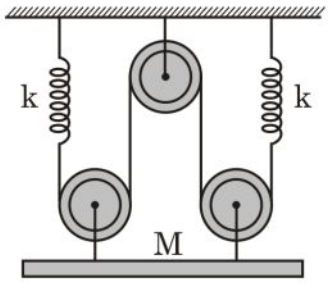
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1. An ammeter and a voltmeter are connected in series to a battery with an emf E = 6V. When a certain resistance is connected in parallel with the voltmeter, the reading of the latter decreases two times, where as the reading of the ammeter increase the same number of times. Find the voltmeter reading, in volt, after the connection of the resistance.

2. An ideal gas having initial pressure p, volume V and temperature T is allowed to expand adiabatically until its volume becomes 5.66 V while its temperature falls to $\frac{T}{2}$. Calculate the degrees of freedom of the gas molecules. Given $\log_e(5.66) = 1.73$, $\log_e(2) = 0.70$

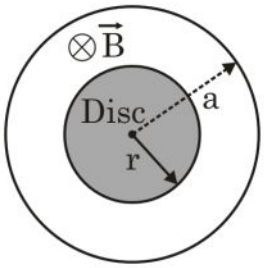
3. The peak emission from a black body at a certain temperature occurs at a wavelength of 6000 Å. On increasing its temperature, the total radiation emitted is increased 16 times. These radiations are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at higher temperature can be bought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level n = 4 to n = 2 in the Bohr's hydrogen atom. The work function of the metal is given by $\frac{\alpha}{100}$eV where α is the numerical constant. The value of α is found 53 × x, find x. [Take : hc = 12420 eV– Å]

4. The natural frequency of the system shown in figure is $f = \frac{1}{\pi}\sqrt{\frac{xk}{M}}$. The pulleys are smooth and massless. Find x.



5. A road passes at some distance from a standing man. A truck is coming on the road with some acceleration. The truck driver blows a whistle of frequency 500 Hz when the line joining the truck and the man makes an angle of θ with the road. The man hears a note having a frequency of 600 Hz when the truck is closest to him. Also, the speed of truck has got doubled during this time. Find the value of θ, in degree.

6. A uniform disc of radius r and mass m is charged uniformly with the charge q. This disc is placed flat on a rough horizontal surface having coefficient of friction μ. A uniform magnetic field is present in a circular region (a > r) but varying as kt^3 as shown in Figure. Find the time, in second after which the disc begins to rotate. (Given r = 1m, m = 18 kg, q = 1 C, $\mu = 0.1$, K = 4, g = 10 ms^{-2})



PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

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Negative Marks	:	–1	In all other cases

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1. $[X] \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} [Y]$

Both [X] and [Y] produce ink blue solution with H_2O_2 in presence of organic solvents and H_2SO_4, [X] & [Y] respectively

(A) MnO_4^-, MnO_4^{-2} (B) $Cr_2O_7^{-2}, CrO_4^{-2}$ (C) $CrO_4^{-2}, Cr_2O_7^{-2}$ (D) SO_3^{-2}, SO_4^{-2}

2. Which of the following set contains, all tetrahedral geometries with respect to central metal atom or ion ?

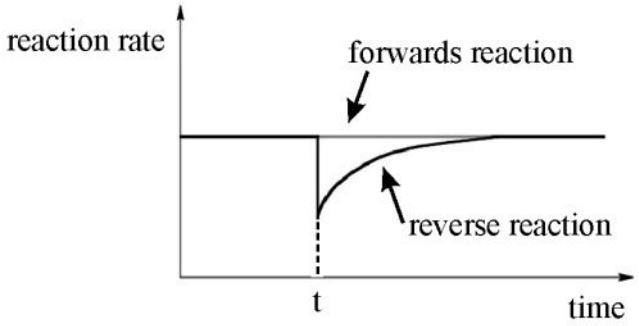
(A) $FeCl_4^{2-}, Cu(CN)_4^{3-}, PdCl_4^{2-}$ (B) $NiCl_2(PPh_3)_2$, $Pt(CN)_4^{2-}, RhCl(PPh_3)_3$

(C) $Ni(CO)_4, Fe(CO)_4^{2-}, Cu(CN)_4^{3-}$ (D) $Fe(CO)_5$, $Co(CO)_4^-$, $[NiCl_4]^{2-}$

3. A small amount of solid lead iodide was added to a beaker of water, which was stirred. Most of the solid settled at the bottom of the beaker, but a little dissolved, establishing the equilibrium

$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2I^-(aq)$

The rates of the forward and reverse reactions were monitored over time, producing the graph shown below:



What happened at time t?

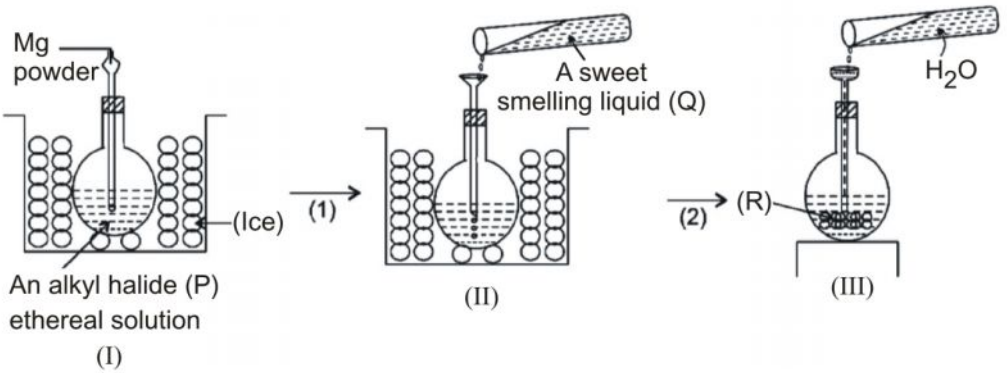
(A) The beaker was cooled in an ice-bath.

(B) A small amount of solid KI was added to the beaker.

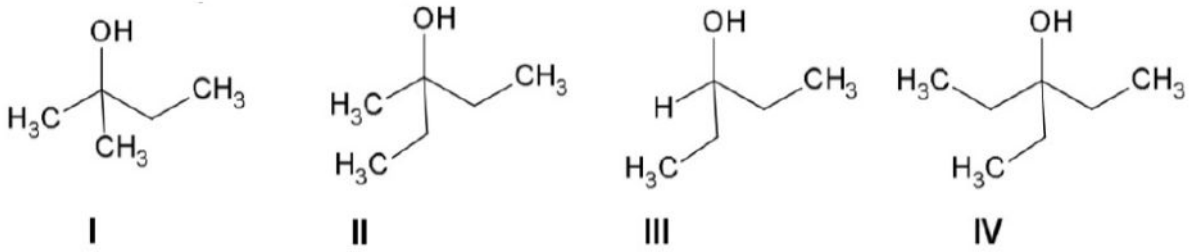
(C) A small amount of solid PbI_2 was removed from the beaker.

(D) A small amount of water was added to the beaker.

4. Consider the following scheme:



If P is ethyl chloride, R can be



(A) I, II, III, IV

(B) II, III, IV only

(C) IV only

(D) II, IV only

SECTION-I (ii) : (Maximum Marks: 12)

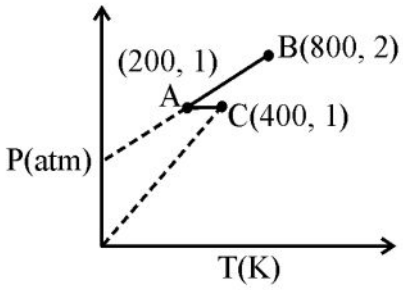
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Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. One mole of a diatomic ideal gas is subjected to a reversible process that involves two steps AB and AC. The pressure at A and C is same. Consider the graph and choose correct statements.
[Given : $\ell n\ 2 = 0.70$, $\ell n\ 3 = 1.1$, R = 0.082 L-atm/K-mol]



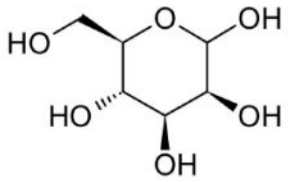
(A) Entropy change for process AC is 0.2 kJK^{-1}

(B) In Path AB work done is –22.3 J

(C) Volume of gas at B = 2 × Volume of gas at A

(D) Volume of gas at C is 32.8 lit.

6. The correct statement/s about the pyranose form of a sugar (X) given below is/are:



(X)

(A) It exists in two anomeric pyranose forms

(B) It reacts with Tollen's reagent to give a silver mirror

(C) The penta-O-methyl derivative of (X) is non reducing

(D) It resists reduction with aqueous sodium borohydride

7. The correct statement(s) related to oxoacids of phosphorous is(are) :

(A) Upon heating, H_3PO_3 undergoes disproportionation reaction to produce H_3PO_4 and PH_3.

(B) While H_3PO_3 can act as reducing agent, H_3PO_4 cannot.

(C) H_3PO_3 is a monobasic acid.

(D) The H atom of P-H bond in H_3PO_3 is not ionizable in water.

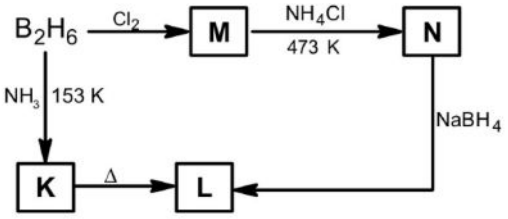
SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases.

---

8. Study the sequence of reactions of diborane (B_2H_6) given below and match the products K, L, M and N with products mention in List-II (shown reagents are taken in excess)



List-I		List-II	
(P)	K	(1)	$[BH_2(NH_3)_2][BH_4]$
(Q)	L	(2)	B_2H_5Cl
(R)	M	(3)	BCl_3
(S)	N	(4)	$B_3N_3H_3Cl_3$
		(5)	$B_3N_3H_6$

(A) P ⟶ 5;Q ⟶ 1;R ⟶ 3;S ⟶ 2

(B) P ⟶ 5;Q ⟶ 1;R ⟶ 2;S ⟶ 4

(C) P ⟶ 1;Q ⟶ 5;R ⟶ 2;S ⟶ 4

(D) P ⟶ 1;Q ⟶ 5;R ⟶ 3;S ⟶ 4

9. Match the half-life of the reaction in List-I with the integrated rate equation in List-II

(a = initial concentration of reactant and a − x − concentration of reactant at time 't'.)

List - I		List - II	
(P)	$t_{1/2} = \frac{0.586}{k}\sqrt{a}$	(1)	$k = \frac{x}{t}$
(Q)	$t_{1/2} = \frac{0.693}{k}$	(2)	$k = \frac{1}{t}\left[\frac{1}{a-x} - \frac{1}{a}\right]$
(R)	$t_{1/2} = \frac{a}{2k}$	(3)	$k = \frac{2}{t}\left[\sqrt{a} - \sqrt{a-x}\right]$
(S)	$t_{1/2} = \frac{1}{ka}$	(4)	$k = \frac{2.303}{t}\log\frac{a}{a-x}$
		(5)	$k = \frac{2}{t}\left[\frac{1}{\sqrt{a-x}} - \frac{1}{\sqrt{a}}\right]$

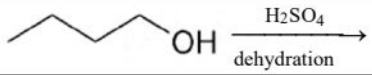
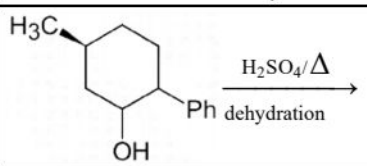
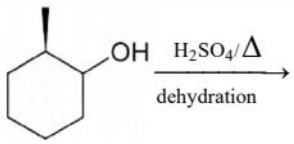
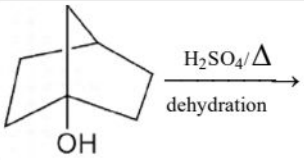
(A) P ⟶ 5;Q ⟶ 4;R ⟶ 1;S ⟶ 2

(B) P ⟶ 3;Q ⟶ 4;R ⟶ 1;S ⟶ 2

(C) P ⟶ 2;Q ⟶ 3;R ⟶ 4;S ⟶ 2

(D) P ⟶ 3;Q ⟶ 2;R ⟶ 4;S ⟶ 5

10. Match each species in List - I with appropriate statement (s) of List - II and choose correct option

List - I		List - II	
(P)		(1)	At least one of the product shows optical isomerism.
(Q)		(2)	Major product contains even number of α - hydrogen's (Hydrogen's which can involve in hyper conjugation).
(R)		(3)	One of the product shows geometrical isomerism.
(S)		(4)	Reactant can turn acidified dichromate to green.
		(5)	Will not undergo dehydration.

(A) P ⟶ 2,3;Q ⟶ 2,4,5;R ⟶ 1,4;S ⟶ 5

(B) P ⟶ 2,3,4;Q ⟶ 2,3,4;R ⟶ 1,2;S ⟶ 3,4

(C) P ⟶ 1,2,3,4;Q ⟶ 2,3,4;R ⟶ 1,3;S ⟶ 2

(D) P ⟶ 2,3,4;Q ⟶ 1,2,3,4;R ⟶ 1,4;S ⟶ 5

11.

List - I		List - II	
(P)	A compound that liberates ammonia on treatment with hot aqueous KOH	(1)	CH_3-CH_2-CN
(Q)	A compound that gives CO_2, NH_4^+ and acetic acid after boiling in concentrated, aqueous HCl	(2)	$CH_3-CH_2-CONH_2$
(R)	A compound that gives NH_4^+ and mono carboxylic acid after treatment with concentrated aqueous HCl.	(3)	$NC-CH_2-CN$
(S)	A compound that gives product A with MeMgI followed by hydrolysis, which on further treatment with MeMgl excess followed by acidification gives 2-methyl-2-butanol	(4)	CH_3COONH_4
		(5)	$PhCOONH_4$

The correct option is_____.

(A) P $\longrightarrow$ 1;Q $\longrightarrow$ 3;R $\longrightarrow$ 5;S $\longrightarrow$ 1

(B) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 3;S $\longrightarrow$ 4

(C) P $\longrightarrow$ 3;Q $\longrightarrow$ 4;R $\longrightarrow$ 5;S $\longrightarrow$ 4

(D) P $\longrightarrow$ 2;Q $\longrightarrow$ 4;R $\longrightarrow$ 2;S $\longrightarrow$ 1

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases.

---

1. Find the number of reactions, which give N_2.

(I) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$ (II) $Ba(N_3)_2 \xrightarrow{\Delta}$

(III) $NH_4Cl \xrightarrow{\Delta}$ (IV) NH_3 (excess) + $Cl_2 \rightarrow$

(V) $NH_4NO_3 \xrightarrow[250^oC]{\Delta}$ (VI) $(NH_4)_2SO_4 \xrightarrow[aq.]{KOH}$

(VII) $C_2H_5NH_2 + HNO_2 \xrightarrow{water}$

2. If 'x' is the total number of $p\pi - d\pi$ bonds and 'y' is the total number of lone pair(s) on non terminal atoms per molecule in trimer of SO_3, the value of $(x + y)$ is :-

3. One gram of commercial $AgNO_3$ is dissolved in 50 mL of water. It is treated with 50 mL of a KI solution. The silver iodide thus precipitated is filtered off. Excess of KI in the filtrate is titrated with M/10 KIO_3 solution in presence of 6 M HC1 till all I^- ions are converted into ICl. It requires 50 mL of M/10 KIO_3 solution. Twenty milliliters of the same stock solution of KI requires 30 mL of M/10 KIO_3 under similar conditions. The percentage of $AgNO_3$ in the sample is (Atomic weight of silver = 108)

4. Both gold and silver are extracted by cyanide process. Gold in nature is frequently alloyed with silver, which is also oxidized by aerated NaCN solution, similar to gold. 1000 L of a solution of 0.002 M $[Au(CN)_2]^-$ and 0.003 M $[Ag(CN)_2]^-$ was evaporated to half of the original volume and treated with 78 g zinc (atomic mass = 65).

Assume that elevation from standard conditions is unimportant and that the redox reactions go to completion. Given:

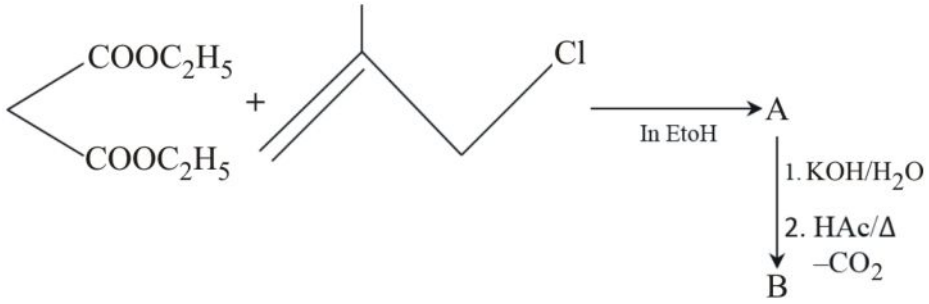
$[Zn(CN)_4]^{2-} + 2e^- \rightarrow Zn(s) + 4CN^-;\ E^\circ = -1.26V$

$[Au(CN)_2]^- + e^- \rightarrow Au(s) + 2CN^-;\ E^\circ = -0.60V$

$[Ag(CN)_2]^- + e^- \rightarrow Ag(s) + 2CN^-;\ E^\circ = -0.31V$

If the concentration of $[Au(CN)_2]^-$ in the solution after addition of zinc in it, be x mol/lit. The value of $(x \times 10^5)$ is.

5. Consider the following reactions



Amount (in mg) of B reacts completely with 100 ml of 0.1 M $NaHCO_3$ solution.

6. The correct order of the given reagents to convert benzene to m-Chlorobromobenzene is

(1) Sulfuric acid (conc.) and heat

(2) Cl_2 + $FeCl_3$ and heat

(3) $NaNO_3$ + H_3O^+, 0°C

(4) H_2/Pt catalyst

(5) Mg in ether

(6) PBr_3

(7) H_3PO_2 (aqueous)

(8) HNO_3 (conc.) + H_2SO_4 (conc.) and heat

(9) Cu_2Br_2 +HBr

(10) $(CH_3CO)_2O$ + pyridine

Write the correct order in form of their number and express last two digits as answer.

[For, e.g. If order is 1357, then your answer will be 57]

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

1. A ray of light travelling along the line OP (O being origin) is reflected by the line mirror $2x - 3y + 1 = 0$, the point of incidence being P(1,1). The reflected ray, travelling along PQ is again reflected by the line mirror $2x - 3y - 1 = 0$, the point of incidence being Q. From Q, ray move along QR, where R lies on the line $2x - 3y + 1 = 0$. Then sum of ordinate of point R and abscissa of point Q is :

(A) $\frac{100}{13}$ (B) $\frac{73}{13}$

(C) $\frac{53}{13}$ (D) $\frac{23}{13}$

2. Number of integers in the range of 'm' for which point (m,1) lies in the smaller region bounded by $x^2 + y^2 - 3x + 1 = 0$ and $2x - y = 2$ is-

(A) 0 (B) 1

(C) 2 (D) 3

3. Let $g(x) = \int_{x^2}^{x^3} \frac{dt}{\ln t}$ for $x > 1$ and $h(x) = \int_{1}^{x} \left(2t^2 - \ln t\right) g(t)dt$ $(x > 1)$, then :

(A) h is increasing on $(1, \infty)$

(B) h is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

(C) h is decreasing on $(1, \infty)$

(D) h is increasing on $(1, 2)$ and decreasing on $(2, \infty)$

4. The tangent at (x', y') to the curve $x^3 + y^3 = a^3$ meets the curve again at (x_1, y_1), then $\left|\frac{x_1}{x'} + \frac{y_1}{y'}\right|^2 =$

(A) 4

(B) 1

(C) a

(D) a^2

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. Let $f(x) = \sin x + px + q$. Then $f(x) = 0$ has :

(A) Only one real root which is positive if $p > 1, q < 0$

(B) Only one real root which is negative if $p < -1, q < 0$

(C) Only one real root which is negative if $p > 1, q > 0$

(D) Only one real root which is positive if $p < -1, q < 0$

6. Let $\vec{u} = \cos\alpha\hat{i} + \sin\alpha\hat{j}$, $\vec{v} = \cos\beta\hat{i} + \sin\beta\hat{j}$, $\vec{w} = \cos\gamma\hat{i} + \sin\gamma\hat{j}$, $0 \le \alpha, \beta, \gamma < 2\pi$. If $\vec{u}.\vec{w} = \vec{v}.\vec{w} = -\sqrt{3}\vec{u}.\vec{v}$ then $\left(2\vec{u}.\vec{v}\right)^3$ is equal to :

(A) $\frac{-8}{27}$ (B) $\frac{-1}{8}$ (C) 1 (D) $\frac{1}{9}$

7. P is a non–singular matrix of order $n \times n$ such that $3PQP^{-1} + P = 2P^{-1}QP$ then

(A) P & Q both are identity matrices

(B) $\det.(P + Q) = 0$

(C) $PQP^{-1} - P^{-1}QP$ is a singular matrix

(D) P + Q is non - invertible matrix

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases.

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8. Matching the following statement

	List-I		**List-II**
(P)	If $\frac{dy}{dx}\left(\frac{1+\cos x}{y}\right) = -\sin x$ and $f\left(\frac{\pi}{2}\right) = -1$ then f(0) is	(1)	5
(Q)	The number of points of non-derivability of the function $f(x) = \left[\frac{2x}{\pi}\right] \operatorname{sgn}\left(\frac{1}{\{x\}}\right)$ in (–2, 2) is (where [P], {P} & sgn(P) denotes greater integer $\leq$ P, fractional part of P & signum function of P respectively)	(2)	–2
(R)	The area bounded by $e^{\ln\lvert x+1\rvert} \geqslant \lvert y\rvert, \lvert x\rvert \leqslant 1$ is	(3)	4
(S)	Number of real values of x which satisfy the equation $\frac{\left(\sec^{-1}x\right)^3 + \left(\operatorname{cosec}^{-1}x\right)^3}{\left(\tan^{-1}x + \cot^{-1}x\right)^3} = 7$ is:	(4)	1
		(5)	7

(A) P ⟶ 1;Q ⟶ 2;R ⟶ 3;S ⟶ 4

(B) P ⟶ 2;Q ⟶ 1;R ⟶ 3;S ⟶ 4

(C) P ⟶ 2;Q ⟶ 3;R ⟶ 1;S ⟶ 4

(D) P ⟶ 2;Q ⟶ 1;R ⟶ 4;S ⟶ 3

9. Matching the following statement

List-I		List-II	
(P)	If $\sin^2 A = x$ and $\sin A \cdot \sin 2A \sin 3A \cdot \sin 4A = ax^2 + bx^3 + cx^4 + dx^5$, then the value of $a + b + c + d + 5$ is	(1)	1
(Q)	The values of x between 0 and 2π which satisfy the equation $\sin x\sqrt{8\cos^2 x} = 1$ are in A.P. with common difference d, then $\frac{4d}{\pi}$ is	(2)	2
(R)	The least difference between the roots, in the first quadrant $\left(0 \leq x \leq \frac{\pi}{2}\right)$, of the equation $4\cos x\left(2 - 3\sin^2 x\right) + (\cos 2x + 1) = 0$ is $\frac{\pi}{k}$, then k =	(3)	6
(S)	Let $f(x) = \sqrt{\sin^4 x + 4\cos^2 x} - \sqrt{\cos^4 x + 4\sin^2 x}$ and $g(\sin 2t) = \sin t + \cos t\ \forall\ t \in \left[\frac{-\pi}{4}, \frac{\pi}{4}\right]$. If the range of g(f(x)) is [a, b], then the value of $a^2 + b^2$ is _____ .	(4)	5
		(5)	4

(A) P ⟶ 2;Q ⟶ 4;R ⟶ 1;S ⟶ 3 (B) P ⟶ 4;Q ⟶ 1;R ⟶ 2;S ⟶ 3

(C) P ⟶ 2;Q ⟶ 1;R ⟶ 4;S ⟶ 3 (D) P ⟶ 4;Q ⟶ 1;R ⟶ 3;S ⟶ 2

10. Matching the following statement

List-I		List-II	
(P)	If $\lim\limits_{h\to 0^+} \int\limits_{-1}^{1} \frac{h\,dx}{h^2 + x^2} = k\pi$ Then k equals	(1)	5
(Q)	The area of the region defined by $1 \leq \lvert x - 2\rvert + \lvert y + 1\rvert \leq 2$ is	(2)	3
(R)	$\int\limits_{0}^{110\pi} [\sin x + \cos x]\,dx = -11k\pi$ then k equal to (where [.] denotes greatest integer function)	(3)	1
(S)	If distance between two focii of ellipse $(x-2y+3)^2 + (8x+4y+4)^2 = 20$ is k then $\frac{k^2}{5}$ equals to	(4)	6
		(5)	9

(A) P ⟶ 3;Q ⟶ 4;R ⟶ 1;S ⟶ 2 (B) P ⟶ 2;Q ⟶ 3;R ⟶ 1;S ⟶ 4

(C) P ⟶ 3;Q ⟶ 4;R ⟶ 2;S ⟶ 1 (D) P ⟶ 2;Q ⟶ 4;R ⟶ 1;S ⟶ 3

11.

List- I		List-II	
(P)	If $\sqrt{x^2+y^2}+\sqrt{(x+3)^2+(y+4)^2}=9-a^2$ represent an ellipse then number of integral value of a is	(1)	7
(Q)	If a, b, c > 0 & $a^2 = bc$ & a + b + c = abc then the least possible value of a^2 is	(2)	5
(R)	The minimum value of $n \in N$ so that the sum of coefficient of terms in the expansion of $(a+b+c-d)^n$ is more than the number of terms is:-	(3)	3
(S)	If $\lim_{x\to\infty}\left(8x^3+ax^2\right)^{\frac{1}{3}}-bx$ exists & is equal to 1, then value of a – 3b is equals to	(4)	6
		(5)	2

(A) P ⟶ 3;Q ⟶ 4;R ⟶ 1;S ⟶ 1

(B) P ⟶ 2;Q ⟶ 1;R ⟶ 4;S ⟶ 1

(C) P ⟶ 2;Q ⟶ 1;R ⟶ 4;S ⟶ 4

(D) P ⟶ 3;Q ⟶ 3;R ⟶ 1;S ⟶ 4

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases.

---

1. The solution of the equation $(2x^2 + 3y^2 - 7)\,x\,dx - (3x^2 + 2y^2 - 8)\,y\,dy = 0$ is given by $(x^2 + y^2 + \gamma) = C(x^2 - y^2 + \delta)^{\beta}$. Where C is arbitrary constant, then $(\beta + \gamma + \delta)$ equals

2. If z is a complex number satisfying the condition $\left|z - \frac{5}{z}\right| = 4$, then sum of greatest and least value of $|z|$ is

3. If T_n is term which is independent of 'x' in the expansion of $\left(\frac{x^4 + x^2 + 1}{x^2 - x + 1} - \frac{x^2 + x + 1}{x}\right)^{12}$, then n is

4. In a drawer Mr. X has 5 pairs of socks, each pair a different color. On Monday, he selects two individual socks at random from the 10 socks in the drawer. On Tuesday, he selects 2 of the remaining 8 socks at random and on Wednesday, two of the remaining 6 socks at random. If the probability that Wednesday is the first day he selects matching socks is $\frac{m}{n}$, where m and n are relatively prime positive integers, then the value of $\frac{m+n}{11}$ is equal to

5. A 3×3 determinant has entries either 1 or -1. Let S be the set of all determinants such that the product of elements of any row or column is -1. For example $\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix}$ is an element of S and number of elements in S is m. Let $P = \begin{vmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix}$ and trace of the matrix adj (adj P) is n then the value of $\frac{m}{n}$ is

6. Let the area of given triangle ABC be Δ. Points A_1, B_1 and C_1 are the mid points of the BC, CA and AB respectively. Point A_2 is the mid point of CA_1. Lines C_1A_1 and AA_2 meet the median BB_1 at points E and D respectively. If Δ_1 be the area of the quadrilateral A_1A_2DE. If the value of $\frac{\Delta_1}{\Delta} = \frac{m}{n}$ (where m and n are coprime), then $n - 3m =$ _____ .

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work



English

(0000CJA101021230019)

Test Pattern

ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
AIOT
12-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $\mathbf{g = 10\ m/s^2}$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

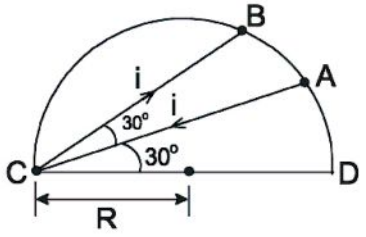
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

---

1. A current carrying wire is placed along ACB in the grooves of an insulating semicircular disc of radius 'R' as shown. The current enters at point A and leaves at point B. Determine the magnetic field at point D.



(A) $\frac{\mu_0 i}{8\pi R\sqrt{3}}$

(B) $\frac{\mu_0 i}{4\pi R\sqrt{3}}$

(C) $\frac{\sqrt{3}\mu_0 i}{4\pi R}$

(D) None

2. In the following radioactive decay C is stable nucleus. Then

$$A \xrightarrow{\lambda_1} B \xrightarrow{\lambda_2} C$$

time t = 0	N_0	0	0
time t	N_1	N_2	N_3

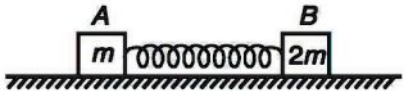
(A) Number of nuclei of B will first increase and then decrease.

(B) Rate of decay of A will first increase and then decrease.

(C) If $\lambda_2 > \lambda_1$ then activity of B will always be higher than activity of A.

(D) If $\lambda_1 > \lambda_2$ then number of nuclei of C will always be less than number of nuclei of B.

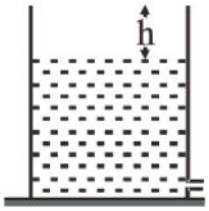
3. In the given figure mass of block A is m and that of B is 2m. They are attached at the ends of a spring and kept on a smooth horizontal surface. Now, the spring is compressed by x_0 and released. Find displacement of block B by the time compression of the spring is reduced to $\frac{x_0}{2}$:-

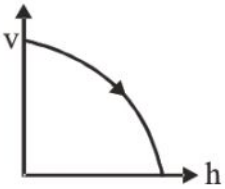


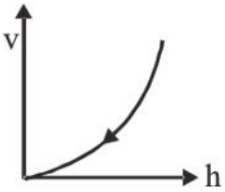
(A) $\left(\frac{x_0}{3}\right)$ (B) $\left(\frac{x_0}{2}\right)$

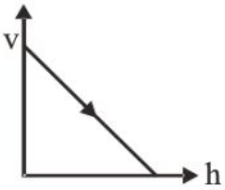
(C) $\left(\frac{x_0}{6}\right)$ (D) $\left(\frac{x_0}{4}\right)$

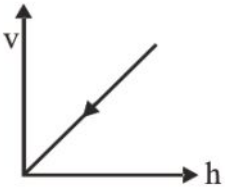
4. A rectangular tank is filled completely with water. A hole at its bottom is unplugged. The graph between the velocity of efflux (through a small hole) vs depth of water h from the top of tank is :-



(A)


(B)


(C)


(D)


SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. The main scale of a vernier calipers reads in mm and its vernier is divided into 10 divisions which coincides with 9 divisions of the main scale. It was also observed when two jaws are brought in contact the zero of the vernier scale coincided with the zero of the main scale. The length of a cube is measured with this vernier calipers. The main scale reads 10 mm and the first division of the vernier scale coincides with the main scale. If the mass of the cube is 2.0606 g then mark the correct alternatives.

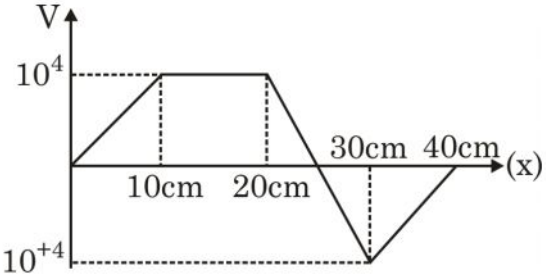
(A) The density of the cube in appropriate significant figures is 2.0000 gm/cm^3.

(B) The density of the cube in appropriate significant figures is 2.00 g/cm^3.

(C) If the zero mark of the vernier scale lies towards the left side of the zero of the main scale when the two jaws are brought in contact, the error will be positive and the correction will be negative.

(D) If the zero mark of the vernier scale lies towards the right side of the zero of the main scale when two jaws are brought in contact the error will be positive and the correction will be negative.

6. For the given potential versus distance graph for one dimensional electric field and a particle of charge 20 μC and mass 0.2 kg, Choose the correct option(s).



(A) Force acting on the particle at x = 25 cm is zero.

(B) Force acting on the particle at x = 25 cm is $4\hat{i}$ Newton.

(C) Minimum speed at origin required by the particle to reach x = 25 cm is 2 m/s.

(D) Minimum speed at origin required by the particle to reach x = 25 cm is $\sqrt{2}$ m/s.

7. A sky lab of mass m is first launched from the surface of earth (radius R) in a circular orbit of radius 2R (from centre of earth) and then it is shifted from this circular orbit to another circular orbit of radius 3R.

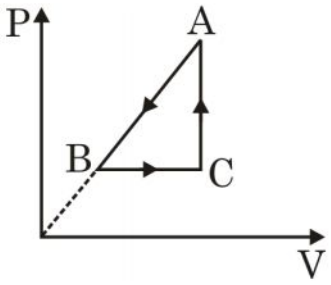
(A) Minimum energy required to place the lab in first orbit is equal to $\frac{3}{4}$ mgR.

(B) Minimum energy required to shift the lab from first orbit to second orbit is $\frac{mgR}{12}$.

(C) Energy of sky lab in second orbit is greater than its energy in first orbit.

(D) Energy of sky lab in second orbit is less than its energy in first orbit.

8. P-V diagram of cyclic process ABCA is as shown in figure. Choose the correct statement(s)



(A) $\Delta Q_{A\to B} = +ve$

(B) $\Delta U_{B\to C} = +ve$

(C) $\Delta U_{CA} = -ve$

(D) $\Delta W_{CAB} = -ve$

9. A ball of mass m moving horizontally at a speed v collides with the bob of a simple pendulum at rest. The mass of the bob is also m. Here $v \leqslant \sqrt{2g\ell}$. (Where ℓ is length of the pendulum)

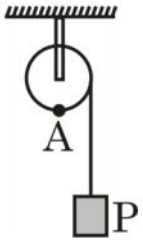
(A) If the collision is perfectly inelastic, the height to which the two balls rise after the collision is $\frac{v^2}{8g}$.

(B) If the collision is perfectly inelastic, the kinetic energy of the system immediately after the collision becomes half of that before collision.

(C) If the collision is perfectly elastic, the bob of the pendulum will rise to a height of $\frac{v^2}{2g}$.

(D) If the collision is perfectly elastic, the kinetic energy of the system immediately after the collision is equal to that before collision.

10. A pulley of radius R = 20 cm is rotated by a weight suspended from a thread which is gradually being wound off the pulley. Initially the weight is at rest and then it begins to go down with an acceleration a = 2 cm/s^2 :-



(A) Angular velocity of the pulley when the weight has travelled a distance S_1 = 100 cm is 1rad/s.

(B) Angular velocity of the pulley when the weight has travelled a distance S_1 = 100 cm is 2rad/s.

(C) The magnitude of the acceleration of point A when the weight has travelled a distance S_1 = 100 cm is $20\sqrt{2}$ cm/s^2.

(D) The magnitude of the acceleration of point A when the weight has travelled a distance S_1 = 100 cm is 20.1 cm/s^2.

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If only the correct answer is given.
Zero Marks	:	0	If no answer is given.
Negative Marks	:	–1	In all other cases

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1. Two thin lenses with lens powers D_1 and D_2 diopter are located at distance L = 25 cm from each other and their optical axes coincide. This system creates an erect real image of the object, located at the optical axis closer to lens D_1 with the net magnification $M_1 = 1$. If the position of the two lenses are exchanged, the system again produces an erect real image with the net magnification $M_2 = 4$. What is the difference between lens powers $\Delta D = D_1 - D_2$ (in diopter)?

2. A storage battery (E = 24 V) was connected to one end of a two conductor telephone line of length L=5.6 km. There is a leakage somewhere in the line. If the free ends are open, then current through the battery is $I_1 = 1.6$ A and if the free ends of the line are short-circuited then current through the battery is $I_2 = 2$ A. The short-circuit current of the battery is $I_3 = 24$ A and resistance of each conductor is $r = 7\Omega$. Determine the resistance (in Ω) of the insulation R at the point of the breakdown.

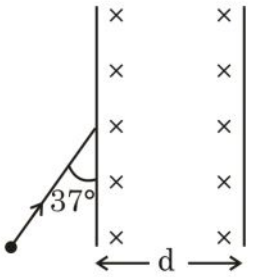
3. A string will break apart if it is placed under too much tensile stress. One type of steel has density $\rho_{steel} = 10^4$ kg/m^3 and breaking stress $\sigma = 9 \times 10^8$ N/m^2. We make a guitar string from (4π) gram of this type of steel. It should be able to withstand (900π) N without breaking. The highest possible fundamental frequency (in Hz) of standing waves on the string is 75n. Find n.

4. A car begins to move at time t = 0 and then accelerates along a straight track with a velocity given by $V(t) = 2t^2$ ms^{-1} for $0 \le t \le 2$, where t is time in second. After the end of acceleration, the car continues to move at a constant speed. A small block initially at rest on the floor of the car begins to slip at t = 1 s and stops slipping at t = 3 s. The coefficient of static and kinetic friction between the block and the floor are μ_s and μ_k respectively. Find the value of $\frac{3\mu_s}{\mu_k}$.

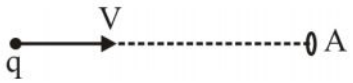
5. A plane spiral with a great number N of turns wound tightly to one another is located in a uniform magnetic field perpendicular to the spiral's plane. The outside radius of the spiral's turns is equal to a and inner radius is zero. The magnetic induction varies with time as $B = B_0 \sin\omega t$, where B_0 and ω are constants. Find the amplitude of emf induced in the spiral. (Given : $\omega = \frac{1}{\pi}$ rad/s, N = 1000, a = 3cm, B_0 = 10 Tesla)

6. A steel rod is 4.000 cm in diameter at 30°C. A brass ring has an interior diameter of 3.992 cm at 30°C. In order that the ring just slides onto the steel rod, the common temperature of the two should be θ (in °C). Find θ/100 rounded off to nearest integer ($\alpha_{steel} = 11 \times 10^{-6}$/°C and $\alpha_{brass} = 19 \times 10^{-6}$/°C)

7. A particle of mass 1 mg and charge 1μC is projected with speed 50 m/s towards a region containing a uniform magnetic field 10 T. What should be the minimum thickness (d) of this region (in m) so that the particle returns back to left side?



8. A point charge is moving along a straight line with constant velocity v. Consider a small area A perpendicular to the direction of motion of charge as shown in figure.



The displacement current through the area when its distance from the charge is x is $\frac{qAv}{2\pi x^n}$. The value of n is :-

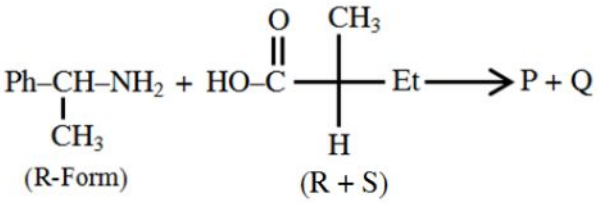
PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

1.



The correct statement regarding product P & Q is :

(A) Both are optically active esters

(B) P & Q are diastereomers

(C) P & Q are meso compounds

(D) Out of P & Q, one is optically active and other is optically inactive.

2. Inversion of sucrose ($C_{12}H_{22}O_{11}$) is first-order reaction and is studied by measuring angle of rotation of the reaction mixture at different instants of time

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$$

If r_0, r_t and r_∞ are the angles of rotation of the reaction mixture at the start, at the time t and at the end of the reaction respectively, then 50% inversion of sucrose occurs when :

(A) $r_0 = 2r_t - r_\infty$

(B) $r_0 = r_t - r_\infty$

(C) $r_0 = r_t - 2r_\infty$

(D) $r_0 = r_t + r_\infty$

3. $\text{CoS (Black)} \xrightarrow{\text{aque-regia}} \text{solution} \xrightarrow[\text{Br}_2 \text{ water}]{\text{exc.NaHCO}_3} \underset{\text{complex (P)}}{\text{coloured}}$

What is co-ordination no. of central metal in complex (P)

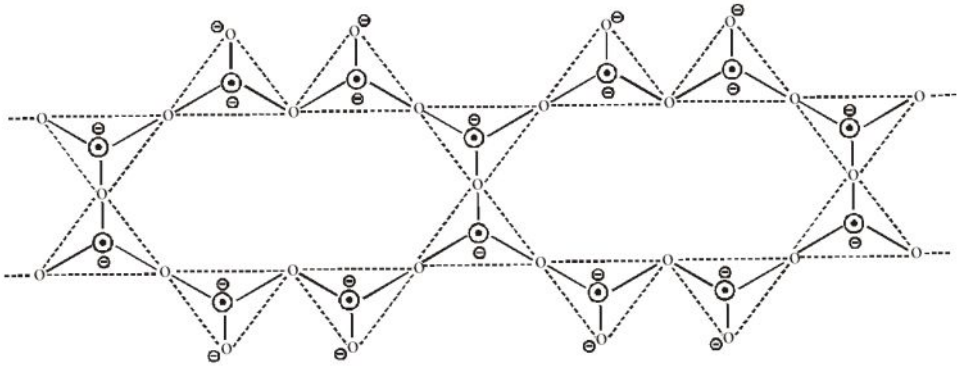
(A) 5

(B) 6

(C) 3

(D) 4

4. What is general formual of below mentioned silicate ? [o : oxygen; • : silicon]



(A) $\left(Si_4O_{11}^{6-}\right)_n$

(B) $\left(Si_2O_5^{2-}\right)_n$

(C) $\left(Si_6O_{17}^{10-}\right)_n$

(D) $\left(Si_3O_8^{4-}\right)_n$

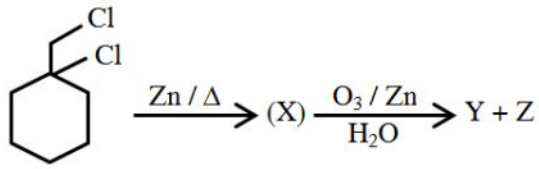
SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

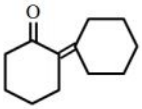
Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

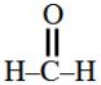
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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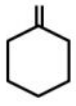
5. 

Product Y and Z are separated and then heated with NaOH. Y gave P where as Z gave Q and CH_3OH as product then which of the following is/are correct :

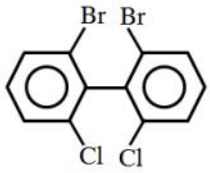
(A) Product P is 

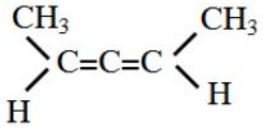
(B) Product Z is 

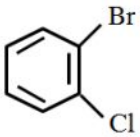
(C) Product Y is 

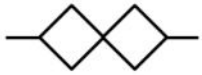
(D) Product X is 

6. Which of the following is/are resolvable ?

(A) 

(B) 

(C) 

(D) 

7. Select the correct statement(s) from the following.

(A) If the critical micelle concentration (CMC) of a soap is 10^{-3} M, then 10^{-4} M solution of this soap will have colloidal nature

(B) Charge on the colloidal particles may be determined with the help of dialysis.

(C) Peptization is the process of conversion of a colloidal sol into precipitate by adding some electrolyte.

(D) When some oil-soluble dye is added in the water in oil emulsion, the background becomes coloured.

8. EMF of the cell: Cd(s) | $CdCl_2.5H_2O$ (sat.) | AgCl(s) | Ag(s) is +0.70 V at 0°C and +0.60 V at 50°C. If $\Delta H°$ and $\Delta S°$ are temperature independent, then the correct information(s) regarding the cell reaction is/are

(A) $\Delta G° = -115.8$ kJ at 50°C

(B) $\Delta G° = 135.1$ kJ at 0°C

(C) $\Delta S° = -386$ J/K

(D) $\Delta H° = -221.178$ kJ

9. Select the correct comparison(s) of properties indicated in options :

(A) $MnO_4^{2-} > MnO_4^-$: (paramagnetic character)

(B) $Na_2Cr_2O_7 > K_2Cr_2O_7$ (Solubility in water)

(C) $Lu^{+3} < Ce^{+3} < La^{+3}$: (Ionic radius)

(D) $Mn^{+3} < Fe^{+3} < Ti^{+3}$: (oxidising power)

10. The extraction of pure metal from bauxite ore (containing iron oxide as chief impurity) involves :

(A) Electrolytic reduction

(B) Poling

(C) Leaching by exc. NaOH(aq.)

(D) Precipitation of amphoteric hydroxide

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If only the correct answer is given.
Zero Marks	:	0	If no answer is given.
Negative Marks	:	–1	In all other cases

---

1. A container of capacity 20 L contains $N_2(g)$, water vapour and negligible but sufficient amount of liquid water to maintain equilibrium, at 300K and 1 atm. The liquid water is now decomposed completely into $H_2(g)$ and $O_2(g)$, by any means, without changing the temperature. If the final pressure becomes 1.9 atm, the mass of liquid water present initially (in gm) was (Nearest integer)
(Aqueous tension of H_2O at 300K = 0.04 atm, R = 0.08 L-atm/K-mole)

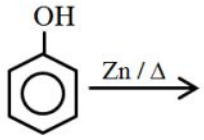
2. Cd^{2+} ion is precipitate after passing H_2S in acidified solution. In the final solution 0.1 milli-mole of $CdSO_4$ are present in 10 ml acidic saturated solution of 0.10M H_2S. What is the pH of the solution after filtering off precipitate, boiling off H_2S and making the solution 100 ml by adding H_2O :
[K_{sp} of CdS = 10^{-28}, For H_2S : $K_{a_1} = 10^{-9}$, $K_{a_2} = 10^{-20}$]

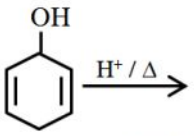
3. Let us assume that de–Broglie equation is valid in Bohr model. In H–like sample, an electron is in an orbit which has path length of the orbit is equal to 3 times of wavelength of electrons in that orbit. This orbit of H–like sample has total energy equal to the ground state energy of H–atom. Now we have same sample having all electrons in the ground state. If photon of energy 126.566 eV is given to this ground state sample then approximate value of wavelength (λ)(in Å, closest integer) of ejected e^- is :

4. Iodine reacts with chlorine in presence of water to give an iodine containing compound 'x'. How many moles of 'x' is formed on reaction of 2 moles I_2 with 10 moles of Cl_2

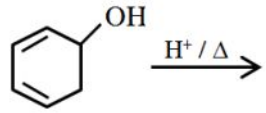
5. A solution contains equimolar amount of following ions : Zn^{+2}, Fe^{+3}, Ag^{+}, Cu^{+2}, Ni^{+2}, Pb^{+2}, Cd^{+2}, Hg^{+2}, Cr^{+3}. How many of cations form soluble complex with excess of aqueus NH_3?

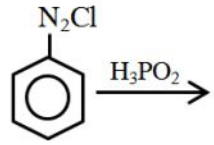
6. How many of the following reactions can give Benzene as final product.

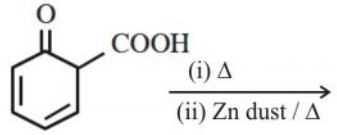
(i) 

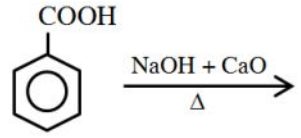
(ii) 

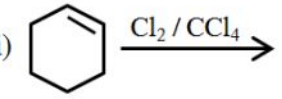
(iii) $CHCl_3 \xrightarrow[\text{ii) Red hot Fe tube}]{\text{i) 6 Ag / } \Delta}$

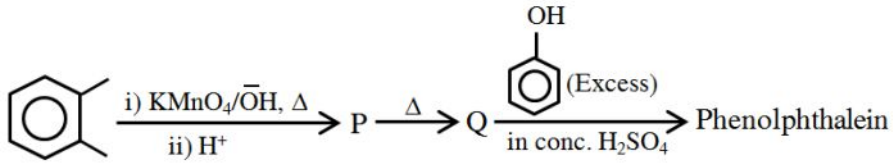
(iv) 

(v) 

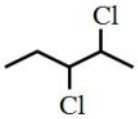
(vi) 

(vii) 

(viii) 

7. 

Find out degree of unsaturation of product Q :

8. How many stereoisomers are possible in 

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

---

1. If x_1 and x_2 are the roots of the equation $\log_{\pi}((\sin^{-1}(x))(\cos^{-1}(x))) = \log_{\sqrt{\pi}}\left(\frac{\pi}{\sqrt{18}}\right)$ for $x \in (0, 1)$ then the value of $\tan^{-1}(x_1^2 + x_2^2)$ is equal to -

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{8}$

2. Let a_n and b_n be the sequences of real numbers such that for all integers $n \geq 0$ where $i = \sqrt{-1}$ and $(3 + i)^n = (a_n + b_n i)$ then the value of $\sum_{n=0}^{\infty} \frac{a_n b_n}{12^n}$ is equal to

(A) $\frac{9}{13}$

(B) $\frac{5}{13}$

(C) $\frac{12}{13}$

(D) $\frac{7}{13}$

3. $\int_0^{\pi/3} \frac{\{\tan x\}}{\tan x}\,dx$ is equal to ({x} denotes fractional part of x)

(A) $\frac{\pi}{3} + \frac{1}{2}\ell n\left(\frac{3}{2}\right)$

(B) $\frac{\pi}{3} - \frac{1}{2}\ell n\left(\frac{3}{4}\right)$

(C) $\frac{\pi}{3} - \frac{1}{2}\ell n\left(\frac{3}{2}\right)$

(D) $\frac{\pi}{3} + \frac{1}{2}\ell n\left(\frac{3}{4}\right)$

4. In a cube ABCDEFGH with opposite vertices C and E, J and I are the mid points of segments FB and HD respectively. Let R be the ratio of the area of the cross section EJCI to the area of one of the faces of the cube then value of R^2 is equal to -

(A) $\frac{5}{4}$

(B) $\frac{3}{2}$

(C) $\frac{4}{3}$

(D) $\frac{25}{16}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. If ${}^{n}C_2 = f(n)$ and $S = {}^{f(3)}C_2 + {}^{f(4)}C_2 + {}^{f(5)}C_2 + \ldots {}^{f(40)}C_2$ then which options are correct for S

(A) Total number of divisors are 144

(B) Total number of odd divisors are 48

(C) S is divisible by 78

(D) Total number of prime factors are 7

6. Let z be a complex number satisfying $12|z|^2 = 2|z + 2|^2 + |z^2 + 1|^2 + 31$ then which options are correct

(A) $\left|z + \frac{6}{z}\right| = 2$

(B) $\text{Re}(z) = -1$

(C) $|z| = \sqrt{6}$

(D) $\frac{3\pi}{4} < \text{Arg}(z) < \pi$

7. For regular tetrahedron ABCD, points E,F are on AB, AC respectively, such that BE= 3, EF = 4, and EF is parallel to plane BCD. Then which options are correct

(A) $DE = \sqrt{37}$

(B) Area of $\Delta DEF = 2\sqrt{33}$

(C) Height of tetrahedron $\sqrt{33}$

(D) Volume of tetrahedron $= \frac{49}{4}\sqrt{\frac{37}{3}}$

8. For triangle ABC, A(0, 8), B(6, 0) and C(x, y) are three points such that C(x, y) lie on circle $(x - 8)^2 + (y - 6)^2 = 9$. If Δ is the area of triangle ABC and z_m & z_M is a point corresponding to complex number of point C for which area is minimum and maximum respectively then select correct options

(A) $\Delta_{max} = 41$

(B) $\Delta_{min} = 11$

(C) $\arg\left(\frac{5(z_M + \bar{z}_M) - 100}{5(z_M - \bar{z}_M) - 78}\right) = -\frac{\pi}{2}$

(D) $\arg(z_m) < \frac{\pi}{4}$

9. Let $f : N \longrightarrow Q$ be a function where N denotes the set of natural numbers and Q denotes the set of rational numbers. $f(1) = \frac{3}{2}$ and $f(x + y) = \left(1 + \frac{y}{x+1}\right) f(x) + \left(1 + \frac{x}{y+1}\right) f(y) + x^2y + xy + xy^2$ for all natural numbers x, y the select correct options :-

(A) $f(20) = 4305$

(B) If $f(n) = 1$ then only one root lie only between (0, 1)

(C) $3^2 + 6^2 + 9^2 + ... + (3n)^2 = 6\, f(n)$

(D) If $2f(n) = (2023)\ (^{n+1}C_2)$ then $n = 1011$

10. Let y(x) be a solution of the differential equation $\left(x - \frac{dy}{dx}\right) \sin x = y\cos x$, $x \in \left(0, \frac{\pi}{2}\right]$. If $y\left(\frac{\pi}{2}\right) = 1$ then choose the correct statement(s) :

(A) y(x) attains a maximum value in $\left(0, \frac{\pi}{2}\right]$

(B) y(x) attains a minimum value in $\left(0, \frac{\pi}{2}\right]$

(C) $\lim\limits_{x \to 0^+} y(x) = 0$

(D) $0 < y\left(\frac{\pi}{4}\right) < 1$

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If only the correct answer is given.
Zero Marks	:	0	If no answer is given.
Negative Marks	:	–1	In all other cases

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1. For parabola $y^2 = 4x$, let P(11, b) be any point inside the parabola and F be the focus of the parabola. If the point Q on the parabola such that FQ + QP is minimum and R is a point of intersection of axis and directrix then the area of parallelogram FPQR is equal to

2. Let A and B are 2×2 matrice such that tr(A) = 7, det(A) = 22, tr(B) = 4, det(B) = 11, tr(AB) = 33. If det(A + xB) = 272 then the value of x is ($x \in I^+$)

3. Let n be a positive integer such that exactly one root of the equation $4x^2 - \left(4\sqrt{3}+4\right)x + \sqrt{3}n - 24 = 0$ is an integer. Then the value of $\frac{n}{2}$ is

4. If $I = \int_0^{\infty} \frac{1}{(1+x^{2024})(1+x^2)}dx$ then the value [I] ([ ] Greatest Integer Function)

5. Let $f : \{1, 2, 3, 4\} \longrightarrow \{a, b, c, d, e, f\}$. If $f(1) \neq a$ or $f(2) \neq b$ or both then number of functions are k then sum of digits of k is

6. A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 compononts randomly from a lot and to accept the lot only if all 3 are non defective. If 30% of the lots have x defective components and 70% have only 1. If purchases reject 46% of lots then value of x is

7. Let T_n consists of general term of positive numbers (where n is positive integer). If $\frac{T_n+2}{2} = \sqrt{2S_n}$ and S_n is sum of n terms, then the value of $(T_{2024} - 8090)$ is equal to

8. Let $f(x) = e^{-|x|}$, $x \in R$ and $g(t) = \int_{-1}^{1} f\left(\frac{x}{t}\right)dx$, $t \neq 0$ then value of $\lim\limits_{t\to 0^+} \frac{g(t)}{t}$ is equal to

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work



ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Advanced)
AIOT
12-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,C	A,B,C	A,D	A,C	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.25	800.00	0.50	1.00	0.50	2.00	0.50	0.20

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C	A,C,D	B,D	D	B,C	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	D	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4.00	5.00	3.00	14.00	0.75	44.00	1170.00	1.50

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,B,C	A,B,C,D	A,D	A,C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.60	2.25	2.25	16.00	14.40	9.66 or 9.67	30.00	2.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C,D)

$I = \frac{V_o}{R}$

$q = C_o V_o - \frac{V_o}{R} t$

$C = \frac{q}{V_o} = C_o \left(1 - \frac{t}{RC_o}\right)$

$\text{Heat} = I^2 R \times \tau$

2. Ans (A,C)

$2mg \sin\alpha = mg\sin\beta$

$\Rightarrow 2\sin\alpha = \sin\beta$

When length of string is maximum

then $\sin\beta = 1$ and $\sin\alpha = \frac{1}{2}$

$\Rightarrow \alpha + \beta = \frac{2\pi}{3}$

$\therefore \ell_{max} = R(\alpha + \beta) = \frac{2\pi R}{3}$

3. Ans (A,B,C)

Conceptual.

4. Ans (A,D)

Point A and C are on the same line passing through origin $\Rightarrow \frac{P_A}{V_A} = \frac{P_C}{V_C}$ ...(1)

Also $T_A = 200\ K = \frac{P_A V_A}{nR}$ and Also $T_C = 1800$

$K = \frac{P_C V_C}{nR} \Rightarrow \frac{P_A V_A}{P_C V_C} = \frac{1}{9}$ ...(2)

From (1) and (2) $\frac{V_A}{V_C} = \frac{1}{3}$

5. Ans (A,C)

From Mosley law,

$\sqrt{\frac{1}{\lambda}} \propto (z-1)$ for K_α line

$\therefore \frac{\lambda_z}{\lambda_1} = \left(\frac{z_{imp_1} - 1}{z_{target} - 1}\right)^2$ ..... (i)

for 2nd impurity $\frac{\lambda_z}{\lambda_2} = \left(\frac{z_{imp_2} - 1}{z_{target} - 1}\right)^2$ ..... (ii)

6. Ans (B,D)

For (B) : Force of buoyancy increases.

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B)

$F = mg\sin\theta + \mu_k mg\cos\theta$ and $p = \vec{F}.\vec{V}$

8. Ans (C)

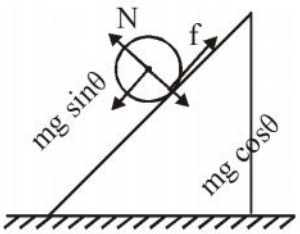
$mg\sin\theta - f = ma$

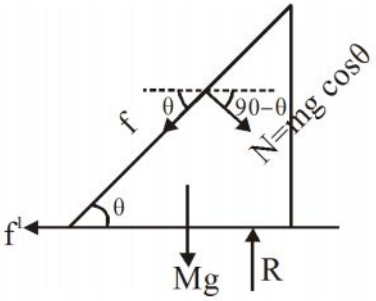
$fR = \frac{MR^2}{2} \cdot \frac{a}{R}$

$mg\sin\theta = \frac{3}{2}ma$

$a = \frac{2g\sin\theta}{3}$

$f = \frac{mg\sin\theta}{3}$

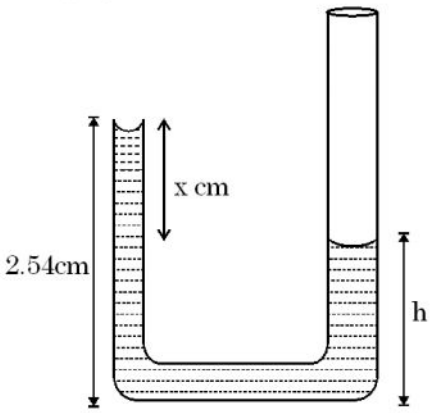




9. Ans (D)

Distant object subtend small angle at eye, so they appear moving along observer.

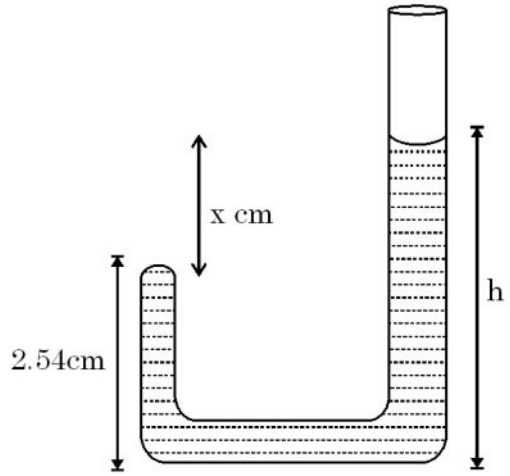
10. Ans (C)



$x = \frac{2T}{\rho g}\left[\frac{2}{d_1} - \frac{2}{d_2}\right] = 1.44$ cm , so h = 1.1 cm

When h > 1.1 cm, radius of meniscus in left limb first increases and then becomes flat for some value of h. For further increase in height, the meniscus in the left limb becomes convex upward and in limiting case the radius of the curvature of the meniscus becomes equal to radius of the capillary.

Under this condition



$P_0 + \frac{4T}{d_1} = P_0 - \frac{4T}{d_2} + x\rho g$

x = 4.32 cm, h = 4.32 + 2.54 = 6.86 cm

PART-1 : PHYSICS

SECTION-II

1. Ans (0.25)

$$I_x = I\frac{x^2}{r^2}, R_x = \frac{i\ell}{\pi x^2}$$

$$\therefore P = \frac{I^2x^2i\ell}{\pi r^4}$$

$$\text{Also } P = -2\pi xK\frac{dT}{dx}$$

$$\Rightarrow \frac{dT}{dx} = -\frac{I^2\rho}{2\pi^2r^4K}x$$

$$\therefore T = T_o + \frac{I^2\rho}{4\pi^2r^2K}$$

2. Ans (800.00)

$$Q = 800 \times 20 = C\Delta T$$

$$\therefore C = \frac{16000}{20} = 800 \text{ J/K}$$

3. Ans (0.50)

$$T = \frac{2m(M-m)g}{M}$$

$$\frac{dT}{dm} = 0 \Rightarrow m = \frac{M}{2}$$

4. Ans (1.00)

$v^2 = I \Rightarrow (v_0 - Ir)^2 = I$

5. Ans (0.50)

$$z_{RL} = \sqrt{R^2 + \omega^2L^2}$$

$$z_{RLC} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\cos \phi_{RL} = \frac{R}{\sqrt{R^2 + \omega^2L^2}}$$

$$\cos \phi_{RLC} = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

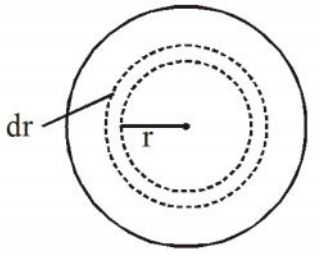
According to question

$$(\omega L)^2 = \left(\omega L - \frac{1}{C\omega}\right)^2 \Rightarrow C = \frac{1}{2\omega^2L}$$

6. Ans (2.00)

$$dm = \frac{M}{\frac{4}{3}\pi R_0^3}.4\pi r^2dr$$

$$dm = \frac{3M}{R_0^3}.r^2dr, \ V = \frac{r}{R_0}V_0$$



$$\text{K. E.} = \frac{1}{2}\int dm\left(\frac{rV_0}{R_0}\right)^2$$

$$= \int \frac{1}{2}\frac{3M}{{R_0}^3}r^2dr.\frac{r^2{V_0}^2}{{R_0}^2}$$

$$= \int \frac{3M{V_0}^2}{2{R_0}^5}\int_0^{R_0} r^4dr = \frac{3M{V_0}^2}{10}, \quad \frac{3M{V_0}^2}{10} = \Delta U$$

$$= \frac{3}{5}GM^2\left[\frac{1}{R_0} - \frac{1}{R}\right]$$

$$R = \frac{2GMR_0}{2GM - R_0{V_0}^2}$$

7. Ans (0.50)

$$\phi = (\vec{E}\cdot\hat{n})A = \left[\frac{(8\times2)+(6\times3)+(10\times6)}{\sqrt{2^2+3^3+6^2}}\right]\left(\frac{7}{47}\right)$$

8. Ans (0.20)

At x = 31.5 mm, both 3600 Å and 6000 Å produce minima. If the slit separation is d, then,

$$\frac{xd}{D\lambda_1} = n_1 + \frac{1}{2}, \ \frac{xd}{D\lambda_2} = n_2 + \frac{1}{2}$$

$$\text{or} \quad \frac{xd}{D}\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) = n_1 - n_2 = 7$$

Substituting the values, D = 1m, λ_1 = 3600Å, λ_2 = 6000 Å

we get d = 0.2 mm.

PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (D)

(P) P = low, $T < T_B$

$\therefore$ $Z < 1$ or $PV \neq nRT$

Gas is more compressible with respect to an ideal gas.

$\because$ $T > T_c$, so gas cannot be liquefied at any pressure at given temperature.

$(V_m)_{real} < (V_m)_{ideal}$

(Q) P = very high, $T = T_B$

$\therefore$ $Z > 1$ or $PV \neq nRT$

$\because$ $T > Tc$, so gas cannot be liquefied at any pressure at given temperature.

$(V_m)_{real} > (V_m)_{ideal}$

(R) P = low, $T = T_B$

$\therefore$ $Z > 1$ or $PV = nRT$

$\because$ $T > Tc$, so gas cannot be liquefied at any pressure at given temperature.

10. Ans (D)

(P) $Ksp = 4.s^3 \Rightarrow s = 10^{-4}\,M \Rightarrow [OH^-] = 2 \times 10^{-4}$

$\therefore$ $pH = 10 + \log 2$

(Q) Basic buffer is formed. $\therefore$ $pH = 14 - 4 - \log\left[\frac{10}{20}\right] = 10 + \log 2$

(R) A salt of weak acid and weak base is formed in which $k_a = k_b$

Hence, pH = 7 or

$$pH = \frac{1}{2}Pk_w + \frac{1}{2}Pk_a - \frac{1}{2}Pk_b$$

(S) $H_3PO_4 + H_2PO_4^-$; Ka_1

$H_2PO_4^{2-} + HPO_4^{2-}$; Ka_2

$HPO_4^{2-} + PO_4^{3-}$; Ka_3

P^H of 0.01 M Na_2HPO_4 =

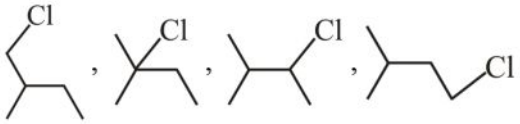
$$\frac{Pk_{a_2} + Pk_{a_3}}{2} = \frac{8.102 + 12.5}{2}$$

$$= \frac{20.602}{2} = 10.301$$

PART-2 : CHEMISTRY

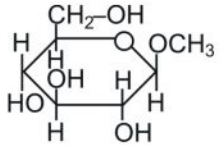
SECTION-II

1. Ans (4.00)



2. Ans (5.00)

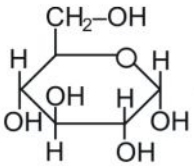
Reducing sugars are those sugars in which anomeric carbon atom has – OH group (Hemiacetal form)



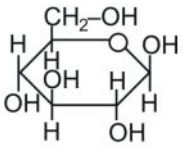
is non reducing sugar (rest all are reducing)

x = 7

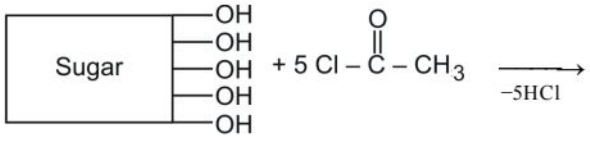
$\rightarrow$ Only these two sugars will be form pentactate because five –OH groups are present

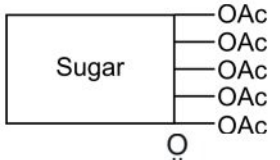


and



y = 2





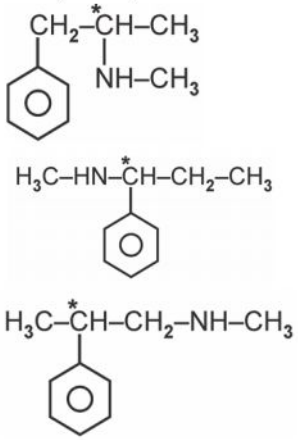
Addition of one $-\overset{O}{\overset{\|}{C}}-CH_3$ group will change in mol. wt. by 42 units

$\therefore$ Number of the $CH_3-\overset{O}{\overset{\|}{C}}-$ groups added

$$= \frac{210}{42} = 5$$

$x - y = 7 - 2 = 5$

3. **Ans (3.00)**



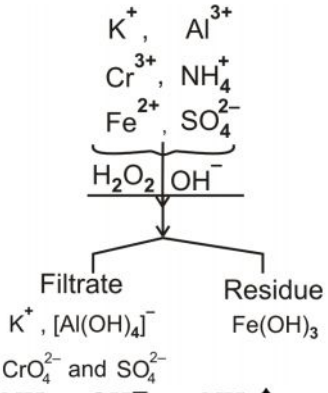
4. **Ans (14.00)**

Potash Alum : $K_2SO_4.Al_2(SO_4)_3.24H_2O$

Chrome Alum : $K_2SO_4.Cr_2(SO_4)_3.24H_2O$

Mohr's salt : $(NH_4)_2.FeSO_4.6H_2O$

In solution :-



CrO_4^{2-} and SO_4^{2-}

$NH_4 + OH^- \longrightarrow NH_3\uparrow$

will be out of system.

$2Fe^{2+} + H_2O_2 \longrightarrow 2Fe^{+3} + 2OH^-$

$Fe^{3+} + 3OH^- \longrightarrow Fe(OH)_3$

Mass of Fe : 56 = a

b = 4 , a/b = 56/4 = 14.00

6. **Ans (44.00)**

	A ⟶	C +	D
t = 0	8	0	0
t = 100 min.	2	6	6

$n_{total} = 2 + 6 + 6 + 4 + 2 = 20$ mole.

$P_T = P_A^oX_A + P_B^oX_B + P_C^oX_C + P_D^oX_D$

$= 90 \times \frac{2}{20} + 40 \times \frac{4}{20} + 50 \times \frac{6}{20} + 40 \times \frac{6}{20}$

$= 9 + 8 + 15 + 12 = \boxed{44 \text{ torr}}$

7. **Ans (1170.00)**

$\Delta H = \Delta U + V\Delta P$

$= 350 + 24.6\left(\frac{4}{3} - 1\right) \times 100$

$= 1170$

8. **Ans (1.50)**

$\overset{+5}{N}O_3^- + 2H^+ + e^- \rightarrow \overset{+4}{N}O_2 + H_2O$; 0.79 volt

$\overset{+5}{N}O_3^- + 8H^+ + 6e^- \rightarrow NH_3OH^+ + 2H_2O$; 0.731 volt

$E = 0.79 - \frac{0.0591}{1}\log\frac{1}{[H^+]^2} = 0.731 - \frac{0.0591}{6}\log\frac{1}{[H^+]^8}$

$0.79 + 2 \times \frac{0.0591}{1}\log[H^+]$

$= 0.731 + \frac{0.0591}{6} \times 8 \times \log[H^+]$

$0.059 = 0.059\left(2 - \frac{8}{6}\right)$ pH

$pH = \frac{6}{4} = 1.5$

PART-3 : MATHEMATICS

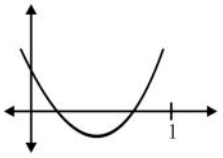
SECTION-I (i)

1. **Ans (A,B,C)**

$f\{x) = ax^2 - bx + c$

S.R. $= \frac{b}{a} < 2 \quad \Rightarrow \quad b < 2a$ ...(1)

P.R. $= \frac{c}{a} < 1 \quad \Rightarrow \quad c < a$ ...(2)



and $D = b^2 - 4ac > 0 \Rightarrow$

$b^2 > 4ac$ ...(3)

To find minimum value of abc

$f(0) = c > 0$; $C_{min.} = 1$

$\therefore \quad (3) \quad \Rightarrow \quad b^2 > 4a$

$(1) \quad \Rightarrow \quad b > 2a$

$\Rightarrow \quad b = 5; a = 5; c = 1$

Also, $f(1) > 0 \Rightarrow a - b + c > 0; \quad a + 1 > b$

$\lambda \le \log_5 25$

$\lambda \le 2$

2. Ans (A,B,C)

$P(A \cap B) = P(A)\,P(B)$

$P(A \cap (B \cap C)) = P(A)P(B \cap C)$

$P(A \cap (B \cup C)) = P(A)\,P(B \cup C)$

$\Rightarrow P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)$

$= P(A)[P(B) + P(C) - P(B \cap C)]$

$\Rightarrow P(A)P(B) + P(A \cap C) - P(A)P(B \cap C)$

$= P(A)P(B) + P(A)P(C) - P(A)P(B \cap C)$

$\Rightarrow P(A \cap C) = P(A)P(C)$

$\Rightarrow$ A and C are independent events.

$\Rightarrow P\left(\frac{A}{C}\right) = P(A) = \frac{1}{2}$

$P\left(\frac{\bar{B} \cup \bar{C}}{A}\right) = P\left(\frac{\overline{B \cap C}}{A}\right) = P\left(\overline{B \cap C}\right)$

$= 1 - P(B)P(C) = 1 - \frac{1}{3}\cdot\frac{1}{4} = \frac{11}{12}$

$P\left(\frac{\bar{A}}{\overline{B \cap C}}\right) = P\left(\bar{A}\right) = \frac{1}{2}$

3. Ans (A,B,C,D)

$\because$ P(3,4) is foot of perpendicular from S (0, 0)

on $3x + 4y - 25 = 0$

which is also on ellipse

$\therefore$ P is vertex of ellipse.

$\therefore$ Distance between focus and directrix = $a - ae$

$= 5$

$\Rightarrow a - \frac{a}{2} = 5 \quad \Rightarrow \quad a = 10$

$\because b^2 = a^2(1 - e^2) = 100\left(1 - \frac{1}{4}\right) = 75$

Clearly AB will be latus rectum of ellipse.

$\therefore$ AB = length of LR = $\frac{2b^2}{a} = 15$

$\therefore$ Focal length = $2ae = 10$

Mid-point of vertex and centre will focus,

$\therefore$ centre = $(-3, -4)$

4. Ans (A,D)

(A) $T_n^3 + T_n + 1 = T_{n+1} - T_n$

$$\sum_{n=1}^{100}\left(T_n^3 + T_n + 1\right) = T_{101} - T_1 + 100 = T_{101} + 99$$

(C) $T_n^2 + 2 = \frac{T_{n+1}}{T_n}$

$$\prod_{n=1}^{100}\left(T_n^2 + 2\right) = \prod_{n=1}^{100}\frac{T_{n+1}}{T_n} = \frac{T_{101}}{T_1} = T_{101}$$

(D) $$\prod_{n=1}^{100}\left(\frac{T_{n+1}}{T_n} - T_n^2\right) = 2^{100}$$

5. Ans (A,C,D)

$f(xy) = e^{xy-x}\cdot f(x) + e^{xy-y} f(y)$

Differentiating w.r.t. x

$$yf'(xy) = e^{xy-x} f'(x) + f(x)e^{xy-x}(y - 1) +$$

$f(y)e^{xy-y}\cdot y$

Put x = 1

$$yf'(y) = e^{y-1} f'(1) + f(1)e^{y-1}(y - 1) + yf(y)$$

Now put x = y = 1 ; f(1) = 0

$\therefore \quad yf'(y) = e^y + yf(y) \quad \Rightarrow \quad x(f'(x) - f(x)) = e^x$

$$e^{-x}(f'(x) - f(x)) = \frac{1}{x}$$

Integrate both sides

$$e^{-x}f(x) = \ell nx + C$$

$f(1) = 0 \qquad \therefore \; C = 0$

$\Rightarrow f(x) = e^x\,\ell nx$. Now verify.

6. Ans (A,B,C,D)

Three planes meet at two points it means they have infinitely many solutions, so

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ \alpha & -1 & 3 \end{vmatrix} = 0$$

$\Rightarrow \quad 2(-3+1) - 1(3+1) + \alpha(1+1) = 0$

$\Rightarrow \quad \alpha = 4$

$P_1 : 2x + y + z = 1$

$P_2 : x - y + z = 2$

$P_3 : 4x - y + 3z = 5$

P on XOY plane $\equiv (1, -1, 0)$

(which can be obtained by putting $z = 0$ in any two of the given planes.)

Q on YOZ plane $\equiv \left(0, \frac{-1}{2}, \frac{3}{2}\right)$

(which can be obtained by putting $x = 0$ in any two of the given planes.)

$\therefore$ Straight line perpendicular to plane P_3 passing through P is :

$$\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$$

$\overrightarrow{PQ} = \hat{i} - \frac{1}{2}\hat{j} - \frac{3}{2}\hat{k}$

Projection of $\overrightarrow{PQ}$ on x-axis $\Rightarrow \left|\frac{\overrightarrow{OP} \cdot \hat{i}}{|\hat{i}|}\right| = 1$

Centroid of ΔOPQ is $\left(\frac{1}{3}, \frac{-1}{2}, \frac{1}{2}\right)$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

(I) 20,b,c are in H.P.

$\Rightarrow \frac{1}{20} + \frac{1}{c} = \frac{2}{b}$

This can be written as $(40 - b)(c + 20) = 800$ we must have $20 < b < 40$ or $0 < 40 - b < 20$,

consider factorisation of 800 in which one term is less that 20

$(40 - b)(c + 20) = 800 = 1 \times 800 = 2 \times 400$

$= 4 \times 200 = 5 \times 160 = 8 \times 100 = 10 \times 80 = 16 \times 50$

Thus pair (b,c) = (39, 780), (38, 380), (36, 180), (35, 140), (32,80), (30,60), (24,30) but be divides c

$\Rightarrow$ Only 5 triples satisfy required requirement

(II) Solution of $m + n + p = 10$, are ${}^{3+10-1}C_{3-1}$

(III) $ab + b - a + 1 = 0$

$\Rightarrow (a + 1)(b - 1) = -2$

$\Rightarrow (a + 1, b - 1) \in \{(1,-2), (-1,2), (2,-1), (-2,1)\}$

$\Rightarrow (a,b) \in \{(0,-1), (-2,3), (1,0), (-3,2)\}$

(IV) Let $z = x + iy$ then $(x + iy)^{20} = x - iy$

$\Rightarrow |z|^{20} = |z| \Rightarrow |z|(|z|^{19} - 1) = 0$

$\Rightarrow |z| = 1$ or 0

$\Rightarrow |z| = 0$

$\therefore$ Ordered pair (0,0)

Let $z \neq 0$ then $(x + iy)^{20}(x + iy) = x^2 + y^2$

$\Rightarrow z^{21} = |z|^2 = 1 \Rightarrow z^{21} - 1 = 0$, which are 21 roots

8. Ans (B)

(I) $f(x) = x^4 + ax^3 + bx^2 + cx + d$

$\alpha, \beta, \gamma, \delta$ are real roots of equation $f(x) = 0$

$\Rightarrow f(x) = (x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$

$\Rightarrow |f(i)| = 1 \Rightarrow (1 + \alpha^2)(1 + \beta^2)(1 + \gamma^2)(1 + \delta^2)$

$= 1$ this is possible if $\alpha = \beta = \gamma = \delta = 0$

$\Rightarrow a = b = c = d = 0$

(II) $x^5 - 1 = 0$

$\Rightarrow (x - 1)(x^4 + x^3 + x^2 + x + 1) = 0$

$$\Rightarrow \frac{x}{1+x^2} + \frac{x^2}{1+x^4} + \frac{x^3}{1+x} + \frac{x^4}{1+x^3}$$

$$= \frac{x}{1+x^2} + \frac{x^2}{1+x^4} + \frac{x^3}{x^5+x} + \frac{x^4}{x^5+x^3}$$

$$= \frac{x}{1+x^2} + \frac{x^2}{1+x^4} + \frac{x^2}{1+x^4} + \frac{x}{1+x^2}$$

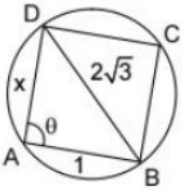
$$= 2\left(\frac{x}{1+x^2} + \frac{x^2}{1+x^4}\right) = \frac{2\left(x + x^5 + x^2 + x^4\right)}{1 + x^2 + x^4 + x^6} = 2$$

(III) $\left[\cos|x| + \frac{1}{2}\right] > 0 \Rightarrow \cos|x| + \frac{1}{2} \geqslant 1$ and

$-\frac{1}{2} \leqslant \cos|x| + \frac{1}{2} \leqslant \frac{3}{2}$

$$\Rightarrow \left[\cos|x| + \frac{1}{2}\right] = 1 \Rightarrow f(x) = 0 \Rightarrow \int_{x_1}^{x_2} f(x) = 0$$

(IV) Let $AD = x$, $\angle BAD = \theta$



In ΔABD, $\frac{BD}{\sin\theta} = 2R$

$$\Rightarrow \sin\theta = \frac{2\sqrt{3}}{4} \Rightarrow \theta = 60°$$

using cosine rule $\cos 60° = \frac{x^2 + 1 - BD^2}{2x}$

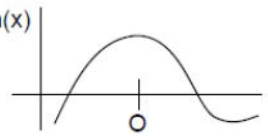
$$\Rightarrow BD = \frac{1 + 3\sqrt{5}}{2}$$

$$\therefore a = \frac{1}{2}, b = \frac{3}{2}$$

9. Ans (A)

(I) $(f(x))^2 + [f'(x)]^2 = h(x)$

$h'(x) = 2f(x)\, f'(x) + 2f'(x)\, f''(x)$



$= 2f'(x)\,(f(x) + f''(x))$

$= 2f'(x)\,\{-x\, g(x)\, f'(x)\}$

$= -2x(f'(x))^2\, g(x)$

so, $h'(x) < 0$ of $x > 0$ and $h'(x) > 0$ for $x < 0$

Then, $x = 0$ is a maxima

(II) $f^2(x) + f(x) = g(x)$

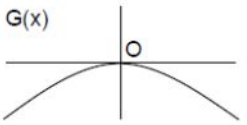
$2f(x)\, f'(x) + f'(x) = g'(x)$ and $g'(x) = f'(x)\{2f(x) + 1\}$

So, $g'(x) \geq 0$ as $g(x)$ is increasing and for $f'(x) \geq 0$

$2f(x) + 1 \geq 0$

Hence, $f(x) \geq -\frac{1}{2}$

(III) Let $G(x) = \left[\int_0^x f(t)\,dt\right]^2$ ....(1)



$$G'(x) = 2f(x)\int_0^x f(t)\,dt$$

and $\frac{G'(x)}{2} = g(x)$

and $G'(0) = 0 = g(0)$

Also, $g(x)$ is non-increasing

Hence, $g(x) \geq 0$

$x \in (-\infty, 0), g(x) \leqslant 0 x \in [0, \infty)$

i.e., $G'(x) \geq 0$ for $x \leq 0$, $G'(x) \leq 0$ for $x \geq 0$ and $G(0) = 0$

But $G(x)$ (from equation (1)) ≥ 0

Hence, $G(x) = 0\ x \in R$

$$\int_0^x f(t)\,dt = 0\ x \in R$$

i.e., $f(x) = 0\ x \in R$

Hence, $f(1) = 0$

(IV) $f(x) = x^2 + \int_0^x e^{-t} f(x-t)\,dt$ ...(1)

$f(x) = x^2 + e^{-x} \int_0^x e^{t} f(x-x+t)\,dt$

$f(x) = x^2 + e^{-x} \int_0^x e^{t} f(t)\,dt$

$f'(x) = 2x + e^{-x}[e^x f(x)] - e^{-x} \int_0^x e^{t} f(t)\,dt$

$f'(x) = 2x + f(x) - \int_0^x e^{-(x-t)} f(t) = 2x + f(x) + x^2 - f(x)$

$f'(x) = 2x + x^2$

$f(x) = x^2 + \frac{x^3}{3} + c$ and from equation (1) $f(0) = 0$

Hence, $c = 0$

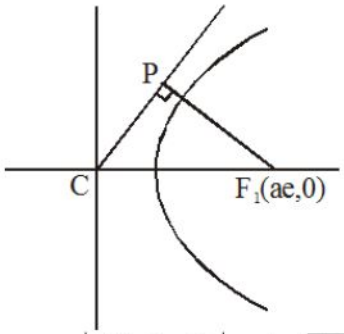
i.e., $f(x) = x^2 + \frac{x^3}{3}$

$f(-1) = 1 - \frac{1}{2} = \frac{2}{3}, \frac{k}{3} = \frac{2}{3} \therefore k = 2$

10. Ans (B)

Length of perpendicular from $F_1(ae, 0)$ upon

$bx - ay = 0$ is equal to



$F_1P = \left|\frac{b(ae) - 0}{\sqrt{a^2 + b^2}}\right| = \frac{b\sqrt{a^2 + b^2}}{\sqrt{a^2 + b^2}} = b$

$\because$ Triangle CF_1P is right angle isosceles

$\therefore$ $CP = F_1P = b$ and $CF_1^2 = CP^2 + PF_1^2$

$\Rightarrow$ $a^2e^2 = 2b^2$ $\Rightarrow$ $a^2e^2 = 2(a^2e^2 - a^2)$

$\Rightarrow$ $e = \sqrt{2}$ $\Rightarrow$ $a = b$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.60)

$$\frac{xf'(g(x))g'(x)}{f(g(x))} = \frac{g'(f(x))f'(x)}{g(f(x))}$$

$$x\frac{d}{dx}(\ell nf(g(x))) = \frac{d}{dx}(\ell ng(f(x)))$$

Also $\int_0^a f(g(x))\,dx = \frac{1}{2} - \frac{e^{-2a}}{2}$ $\forall$ a

$\therefore$ On differentiate, we get $f(g(a)) = e^{-2a} \Rightarrow f(g(x))$

$= e^{-2x}$

From Eqns. (1) and (2), we get

$$x\frac{d}{dx}(\ell ne^{-2x}) = \frac{d}{dx}(\ell ng(f(x)))$$

$$-2x = \frac{d}{dx}(\ell ng(f(x)))$$

In integrating, we get

$$\ell ng(f(x)) = -x^2 + C$$

$$g(f(x)) = k \cdot e^{-x^2}$$

Given $g(f(0)) = 1$ $\Rightarrow$ $k = 1$

$\therefore$ $g(f(4)) = e^{-16}$ $\Rightarrow$ $\lambda = 16$

$$\frac{\lambda}{10} = 1.60$$

2. Ans (2.25)

Centre of circle = $(4, -5)$

Radius = $\sqrt{|-4 + 5i|^2 - (-40)} = \sqrt{4^2 + 5^2 + 40} = 9$

Distance of centre $(4, -5)$ from $(-2, 3)$ is 10.

$\therefore$ $a = \max.|z - (-2 + 3i)| = 9 + 10 = 19$

$b = \min.|z - (-2 + 3i)| = 10 - 9 = 1$

$$\frac{a - b}{8} = \frac{18}{8} = \frac{9}{4} = 2.25$$

3. Ans (2.25)

$f(1) + g(1) = 9e$; $f(x) = -x^2 g'(x)$ and $g(x)$

$= -x^2 f'(x)$

$$I = \int_1^4 \frac{f(x)+g(x)}{x^2}\,dx = \int_1^4 \frac{-x^2(f'(x)+g'(x))}{x^2}\,dx$$

$$= \int_1^4 -\left(f'(x)+g'(x)\right)dx$$

$= -[f(x) + g(x)]_1^4 = -(f(4) + g(4) - 9e)$

Now, $f(x) + g(x) = -x^2(g'(x) + f'(x))$

$$\Rightarrow \frac{g'(x)+f'(x)}{f(x)+g(x)} = \frac{-1}{x^2} \Rightarrow f(1) + g(1) = 9e$$

$\Rightarrow a = 9$

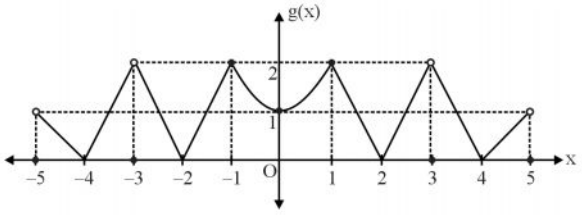
$\Rightarrow f(x) + g(x) = 9e^{1/x}$

$I = -(9e^{1/4} - 9e) = 9(e - e^{1/4}) \equiv k(e - e^{1/4})$

$\Rightarrow k = 9$

$$\frac{k}{4} = \frac{9}{4} = 2.25$$

4. Ans (16.00)



$g(g(x)) = 1$, let $g(x) = t$

$g(t) = 1 \Rightarrow t = 0, t_1, t_2, t_3, t_4, t_5, t_6$

Where $1 < t_1 < 2, 2 < t_2 < 3, 3 < t_3 < 4, t_4, t_5, t_6 < 0$

Now, $\left.\begin{matrix} g(x) = 0 & \Rightarrow & 8 \text{ solutions} \\ g(x) = t_1 & \Rightarrow & 8 \text{ solutions} \end{matrix}\right\} \Rightarrow 16$ solutions

5. Ans (14.40)

$\det.(A^2 + A) = \det.[(P^{-1}DP)(P^{-1}DP) + P^{-1}DP]$

or $\det.(P^{-1}D^2P + P^{-1}DP)$

or $\det.(P^{-1}(D^2 + D)P)$

or $\det.(P^{-1}P)\cdot\det.(D^2 + D)$

or $\det.(I)\cdot\det.(D)\cdot\det.(D + 1)$

or $1^3\cdot(1\cdot2\cdot3)\cdot(2\cdot3\cdot4) = 6\cdot24 = 144$

6. Ans (9.66 or 9.67)

$$d = \frac{-57 - 303}{30} = -12$$

$$A_r = 303 + r \times (-12) \geq 0 \Rightarrow r \leqslant \frac{303}{12} \simeq 25...$$

$A_{25} = 303 + 25(-12) = 3$

$|A_r|_{min.} = |A_{25}| = 3$

$$\frac{S}{(A_{14} - 12)|A_r|_{min.}} = \frac{29(A_1 + A_{29})}{2(A_{15})\cdot 3} = \frac{29}{3}$$

7. Ans (30.00)

$\left(x - 2 + \frac{1}{x}\right)^{30} = n_0x^{30} + n_1x^{29} + ... + n_{29}x + n_{30} + n_{31}x^{-1} + ... + n_{60}x^{-30}$

$(x - 1)^{60} = n_0x^{60} + n_1x^{59} + ... + n_{29}x^{31} + n_{30}x^{30} + n_{31}x^{29} + ...n_{60}$

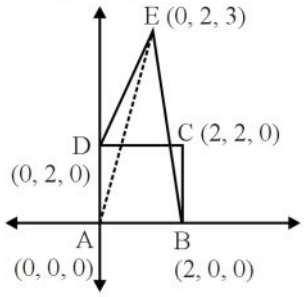
$\therefore n_0 = {}^{60}C_0,\ n_1 = -{}^{60}C_1,\ n_2 = {}^{60}C_2,\ ...,\ n_{30} = {}^{60}C_{30} \ldots n_{60} = {}^{60}C_{60}$

$\therefore C = n_0 + n_1 + n_2 + ... + n_{60} = 0$

Hence, $C - n_{30} = -{}^{60}C_{30}$

$\therefore a = 30.00$

8. Ans (2.00)



$$V = \frac{1}{6}\left[\overrightarrow{AB}\quad\overrightarrow{AD}\quad\overrightarrow{AE}\right] = 2$$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Advanced)
AIOT
12-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	C	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,D	B,D	A,B,C	B,D	A,B,C,D	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	6	5	4	3	3	1	3

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	B	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C,D	A,B,D	D	A,C	A,B,C	A,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	9	2	6	4	5	7	7	4

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	C	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C	A,B,C	A,B,C,D	A,B,D	A,C,D	B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	6	5	6	0	9	4	4	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

$\because r_1 = 2R\sin 30^\circ,\ r_2 = 2R\sin 60^\circ = R\sqrt{3} = R$

$$B_D = B_{AC} + B_{CB} = \left(\frac{\mu_0 i}{4\pi r_1}\sin 60^\circ - \frac{\mu_0 i}{4\pi r_2}\sin 30^\circ\right)$$

$$\Rightarrow B_D = \frac{\mu_0 i}{4\pi R\sqrt{3}}$$

4. Ans (A)

$v = \sqrt{2g(H-h)}$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,D)

The observed reading will be = 10 mm + 1 × 01 mm = 1.01 cm

$$\therefore \text{ density} = \frac{\text{Mass}}{\text{vol}} = \frac{2.0606}{(1.01)^3} = 1.9999$$

$\approx 2.00\ \text{gm/cm}^3$

6. Ans (B,D)

$$F = qE = -q\frac{dv}{dx}$$

$$\frac{1}{2}mv_0^2 = q(10^4)$$

7. Ans (A,B,C)

$E_S = \frac{-GMm}{R}; E_1 = \frac{-GMm}{2(2R)}; E_2 = \frac{-GMm}{2(3R)}$

$E_1 - E_S = \frac{3}{4}mgR$

$E_2 - E_1 = \frac{mgR}{12}$

8. Ans (B,D)

(B) as temperature increase

(D) as work is done on gas

9. Ans (A,B,C,D)

Using conservation of linear momentum and conservation of energy

PART-1 : PHYSICS

SECTION-III

1. Ans (3)

If x is the distance of object from the first lens

So, $m_1 = \frac{f_1}{f_1 - x}$ where f_1 is focal length of first lens

and $m_2 = \frac{f_2}{f_2 - \left(L - \frac{xf_1}{f_1 - x}\right)}$ where f_2 is focal length of the second lens and L is separation between the lenses.

So, $\frac{1}{M_1} = \frac{1}{m_1 m_2} = \frac{f_1 - x}{f_1} \frac{f_2 - \left(L - \frac{xf_1}{x - f_1}\right)}{f_2}$

$= 1 - \frac{L}{f_2} - \frac{x(f_1 + f_2 - L)}{f_1 f_2}$

When the lenses are interchanged, in the above expression only the second term changes

So, $\frac{1}{M_2} = 1 - \frac{L}{f_1} - \frac{x(f_1 + f_2 - L)}{f_1 f_2}$

$\therefore \frac{1}{M_1} - \frac{1}{M_2} = L\left(\frac{1}{f_1} - \frac{1}{f_2}\right) = L(D_1 - D_2)$

So, $1 - \frac{1}{4} = \frac{1}{4}(D_1 - D_2)$

$\therefore D_1 - D_2 = 3$ diopter

2. Ans (6)

$r_{cell} = \frac{24}{24} = 1\Omega$

$\frac{24}{1.6} = 15 = 2x + R + 1$

$\frac{24}{2} = 12 = 2x + 1 + \frac{R(14 - 2x)}{R + (14 - 2x)}$

solving $R = 6\Omega$

3. Ans (5)

$f = \frac{1}{2l}\sqrt{\frac{T}{\rho A}}$ .....(1)

$\frac{m}{Al} = \rho$ .....(2)

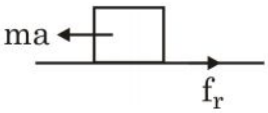
$\sigma = \frac{T}{A}$ .....(3)

$\therefore f = \frac{T}{2m}\sqrt{\frac{\rho}{\sigma}} = 375$ Hz

4. Ans (4)

$V = 2t^2$

$a = 4t$



at t = 1 sec slipping occure

$ma = \mu_s mg$

$4t = \mu_s g$

$\mu_s = 0.4$

$t = 1 \longrightarrow t = 3$ slipping occur

b/w t = 1 and t = 2sec.

$(4t - \mu_k g) = \frac{dv}{dt}$

$v = \left[2t^2 - \mu_k gt\right]_1^2 v = 2 \times 3 - \mu_k g$

$\mu_k = 0.3 \frac{3\mu_s}{\mu_k} = 4$

5. Ans (3)

$\varepsilon_{im} = \frac{1}{3}\pi a^2 N \omega B_0 = 3$

6. **Ans (3)**

For ring just slides on to the steel rod the diameter of rod and ring should be equal to each other and suppose due to AG increment in temperature the diameter of both are equal then

$4(1+\alpha_S\,\Delta\theta) = 3.992\,(1+\alpha_{Brass}\,\Delta\theta)$

$4 + 4\times 11\times 10^{-6}\times\Delta\theta = 3.992 + 3.992\times 20\times 10^{-6}\times\Delta\theta$

$4 + 44\times 10^{-6}\,\Delta\theta = 3.992 + 79.84\times 10^{-6}\times\Delta\theta$

$0.008 = 35.84\times 10^{-6}\,\Delta\theta$

$\frac{8\times 10^3}{35.84} = \Delta\theta\ ;\ \Delta\theta = \frac{8000}{35.84} = 283$

so if temperature increased by 223°C then ring will start to slide and this temperature will equal to

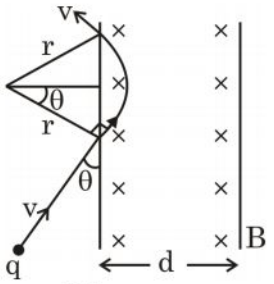
$\theta = 30° + \Delta\theta = 30 + 253 = 283°C$

$\theta = 283°C \approx 280°C$

7. **Ans (1)**

For particle to came back to left side,

$d > (r - r\cos\theta)$



$d > \frac{mv}{qB}(1-\cos\theta)$

$m = 1\ mg = 10^{-6}\ kg$

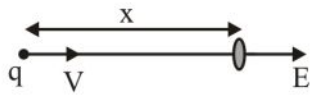
$v = 50\ m/s,\ q = 10^{-6}\ C,\ B = 10\ T$

$\cos\theta = 4/5$

$\Rightarrow d > \frac{10^{-6}\times 50}{10^{-6}\times 10}\left(1-\frac{4}{5}\right)$

$\Rightarrow\ d > 1m$

8. **Ans (3)**



$\phi_E = E.A = \left(\frac{1}{4\pi\epsilon_0}\right)\frac{q}{x^2}.A$

$I_d = \left|\epsilon_0\cdot\frac{d\phi_E}{dt}\right|$

$= \left|\epsilon_0\cdot\frac{qA}{4\pi\epsilon_0}\frac{d}{dt}\left(x^2\right)\right| = \frac{qAv}{2\pi x^3}$

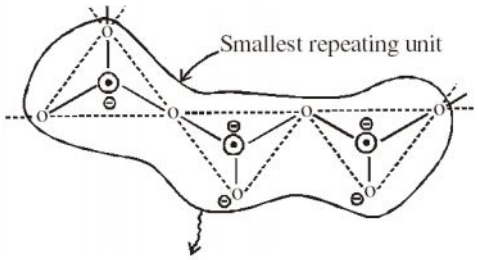
PART-2 : CHEMISTRY

SECTION-I (i)

2. **Ans (A)**

$\frac{\ell n\,2}{K} = \frac{1}{K}\ell n\left[\frac{r_\infty - r_0}{r_\infty - r_t}\right]$

4. **Ans (C)**

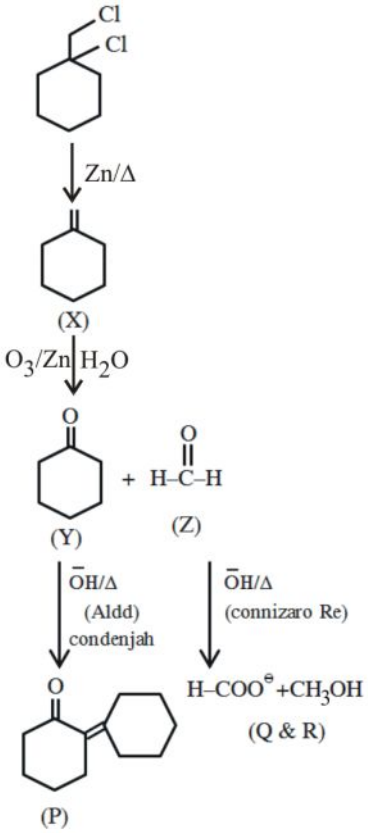


$Si_3O^{5-}_{7+3\left(\frac{1}{2}\right)} = Si_3O^{5-}_{\frac{17}{2}} \equiv \left(Si_6O^{10-}_{17}\right)_n$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A,B,C,D)



7. Ans (D)

Below CMC, the solution is true solution.

8. Ans (A,C)

Net cell reaction is

$$Cd(s) + 2AgCl(s) \rightarrow 2Ag(s) + Cd^{2+}_{(aq)}(aq) + 2Cl^-$$

$\Delta G^\circ_{50^\circ C} = -nFE^\circ = -2 \times 96500 \times 0.6 = -115800$ J

$\Delta G^\circ_{0^\circ C} = -nFE^\circ = -2 \times 96500 \times 0.7 = -135100$ J

$$\Delta S^\circ = nF.\left(\frac{E_2^\circ - E_1^\circ}{T_2 - T_1}\right) 2 \times 96500 \times \frac{0.6 - 0.7}{50}$$

$= -386$ J/K

$\Delta H^\circ = \Delta G^\circ + T\,.\,\Delta S^\circ$

$= (-135100) + 273 \times (-386) = 240478$ J

9. Ans (A,B,C)

Oxidising strength = $Ti^{+3} < Fe^{+3} < Mn^{+3}$

PART-2 : CHEMISTRY

SECTION-III

1. Ans (9)

$P_{N_2} = 1 - 0.04 = 0.96$ atm

$$H_2O(\ell) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$$

Initial	x mole	0	0
Final	0	x	x/2

Now,

$$\frac{3x}{2} = \frac{(1.9 - 0.96 - 0.04) \times 20}{0.08 \times 300}$$

$\Rightarrow x = 0.5$

$\therefore$ Mass of liquid water present initially

$= x \times 18 = 9$ gm

2. Ans (2)

$$[Cd^{2+}] = \frac{0.1 \times 10^{-3}}{10} \times 1000 = 10^{-2}M$$

$$[S^{2-}] = \frac{10^{-28}}{10^{-2}} = 10^{-26}$$

$$10^{-29} = \frac{10^{-26}[H^+]^2}{0.1}$$

$[H^+] = 10^{-2}$

3. Ans (6)

$2\pi r = 3\pi \Rightarrow \frac{2\pi r}{\lambda} = 3 = n$

T.E. in the orbit = -13.6 eV

$\Rightarrow -\frac{13.6\,Z^2}{3^2} = -13.6 \Rightarrow Z = 3$, Sample is Li^{2+}.

Ground state energy = -13.6×9 eV

Energy of ejected e^- = 126.566 – 122.4 = 4.166

$$\lambda = \left(\frac{150}{4.16}\right)^{\frac{1}{2}} = 6\text{Å}$$

4. Ans (4)

$I_2 + 6H_2O + 5Cl_2 \rightarrow 2HIO_3 + 10HCl$

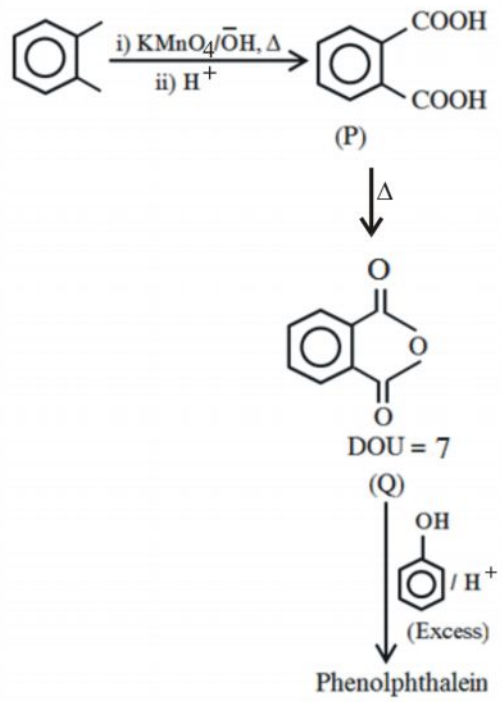
5. Ans (5)

Zn^{+2}, Ag^+, Cu^{+2}, Cd^{+2}, Cr^{+3} , Ni^{+2}

6. Ans (7)

i, ii, iii, iv, v, vi, vii

7. Ans (7)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$\log_{\pi}(\sin^{-1}(x)\,(\cos^{-1}(x)) = \log\left(\frac{\pi^2}{18}\right)$$

$$\Rightarrow \sin^{-1}(x)\left(\frac{\pi}{2} - \sin^{-1}(x)\right) = \frac{\pi^2}{18}$$

$$\Rightarrow (\sin^{-1}(x))2 - \frac{\pi}{2}(\sin^{-1}(x)) + \frac{\pi^2}{18} = 0$$

$$\Rightarrow \left(\sin^{-1}(x) - \frac{\pi}{3}\right)\left(\sin^{-1}(x) - \frac{\pi}{6}\right) = 0$$

$$\Rightarrow \sin^{-1}(x) = \frac{\pi}{3} \text{ or } \frac{\pi}{6}$$

$$x = \frac{\sqrt{3}}{2} \text{ or } \frac{1}{2}$$

$$x_1^2 + x_2^2 = \frac{3}{4} + \frac{1}{4} = 1$$

$$\text{So } \tan^{-1}(1) = \frac{\pi}{4}$$

2. Ans (A)

$$(3 + i)^n = (a_n + b_n i)$$

squaring both side

$$(8 + 6i)^n = a_n^2 - b_n^2 + 2a_n b_n i$$

$$\Rightarrow \left(\frac{8+6i}{12}\right)^n = \frac{a_n^2 - b_n^2}{12^n} + \frac{2a_n b_n}{12^n} i$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{a_n b_n}{12^n} = \frac{1}{2}\sum_{n=0}^{\infty} \text{Im}\left(\frac{8+6i}{12}\right)^n$$

$$= \frac{1}{2}\left[\frac{1}{1 - \left(\frac{4+3i}{6}\right)}\right]$$

$$= \frac{1}{2}\left[\frac{6}{2-3i}\right] = \frac{1}{2}6\left[\frac{2+3i}{13}\right] = \frac{9}{13}$$

3. Ans (C)

$$\Rightarrow \int_0^{\frac{\pi}{4}} \frac{\tan x - [\tan x]}{\tan x} dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan x - [\tan x]}{\tan x} dx$$

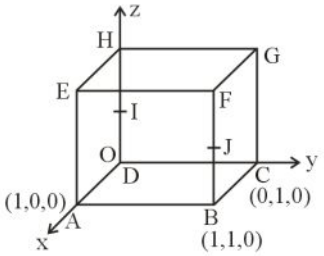
$$\Rightarrow \int_0^{\frac{\pi}{4}} 1\, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} 1 - (\cot x)\, dx$$

$$= \frac{\pi}{4} + \left(\frac{\pi}{3} - \frac{\pi}{4}\right) - (\ell n \sin x)_{\pi/4}^{\pi/3}$$

$$= \frac{\pi}{3} - \left(\ell n \frac{\sqrt{3}}{2} - \ell n \frac{1}{\sqrt{2}}\right)$$

$$= \frac{\pi}{3} - \frac{1}{2}\ell n\left(\frac{3}{2}\right)$$

4. **Ans (B)**



$E(1, 0, 1),\ I\left(0, 0, \frac{1}{2}\right),$

$J\left(1, 1, \frac{1}{2}\right),\ C(0, 1, 0)$

$\Delta EIJ + \Delta IJC$

$\overrightarrow{EI} = \left(1, 0, \frac{1}{2}\right) \quad \overrightarrow{EJ} = \left(0, -1, \frac{1}{2}\right)$

$$\Delta = \frac{1}{2}\left|\overrightarrow{EI} \times \overrightarrow{EJ}\right|$$

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} \end{vmatrix}$$

$$= \frac{1}{2}\left|\hat{i}\left(\frac{1}{2}\right) - \hat{j}\left(\frac{1}{2}\right) + \hat{k}(-1)\right|$$

$$= \frac{1}{2}\sqrt{\frac{1}{4} + \frac{1}{4}} + 1 = \frac{1}{2}\sqrt{\frac{3}{2}}$$

$\Rightarrow$ Total surface area $= \frac{\sqrt{3}}{2}$

$\Rightarrow R = \sqrt{\frac{3}{2}} \quad R^2 = \frac{3}{2}$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,C)**

$$\sum_{i=3}^{40} \begin{pmatrix} \begin{pmatrix} i \\ 2 \end{pmatrix} \\ 2 \end{pmatrix} = \sum_{i=3}^{40} \begin{pmatrix} \frac{i(i-1)}{2} \\ 2 \end{pmatrix}$$

$$= \sum_{i=3}^{40} \frac{\frac{i(i-1)}{2}\left(\frac{i(i-1)}{2} - 1\right)}{2}$$

$$= \frac{1}{8}\sum_{i=3}^{40} i(i-1)(i(i-1)-2)$$

$$= \frac{1}{8}\sum_{i=3}^{40} i(i-1)\left(i^2 - i - 2\right)$$

$$= \frac{1}{8}\sum_{i=3}^{40} i(i-1)(i+1)(i-2)$$

$$= \frac{1}{8}\sum_{i=3}^{40} i(i-2)(i-1)i(i+1)$$

$$= \frac{1}{40}\sum [(i-2)(i-1)i(i+1)(i+2) - (i-3)(i-2)(i-1)i(i+1)]$$

$$= \frac{38.39.40.41.42 - 0}{40}$$

$= 38 \cdot 39 \cdot 41 \cdot 42$

$= 19 \times 2^2 \times 13 \times 3 \times 41 \times 7 \times 3$

$S = 2^2 \times 3^2 \times 7 \times 13 \times 19 \times 41$

6. **Ans (A,B,C)**

$12|z|^2 = 2|z + 2|^2 + |z^2 + 1|^2 + 31$

$\Rightarrow 12z\bar{z} = 2(z + 2)(\bar{z} + 2) + \left(z^2 + 1\right)\left(\bar{z}^2 + 1\right) + 31$

$\Rightarrow 12z\bar{z} = 2(z\bar{z} + 2(z + \bar{z}) + 4) + z^2\bar{z}^2 + z^2 + \bar{z}^2 + 1 + 31$

$\Rightarrow \left[\left(z^2 + 2z\bar{z} + \bar{z}^2\right) + 4(z + \bar{z}) + 4\right] + z^2\bar{z}^2 - 12z\bar{z} + 36 = 0$

$\Rightarrow (z + \bar{z} + 2)^2 + (z\bar{z} - 6)^2 = 0$

$\Rightarrow z + \frac{6}{z} = z + \bar{z} = -2$

7. Ans (A,B,C,D)

Clearly, EF is parallel to BC.

Sine all the faces of ABCD are equilateral triangles,

we have

$AE = AF = EF = 4$, $AD = AB = AE + BE = 7$

By the law of cosines,

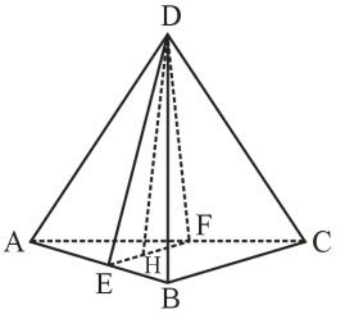
$DE = \sqrt{AD^2 + AE^2 - 2AD.\,AE.\cos 60^\circ}$

$= \sqrt{49 + 16 - 28} = \sqrt{37}$,

and $DF = \sqrt{37}$ likewise.

Let DH be an altitude of isosceles ΔDEF,

H being the foot on EF, as shown in the figure

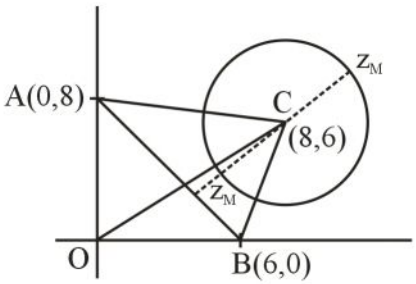


Then $EH = \frac{1}{2}EF = 2$, and

$DH = \sqrt{DE^2 - EH^2} = \sqrt{33}$

We conclude $S_{\Delta DEF} = \frac{1}{2}$. EF . DH $= 2\sqrt{33}$.

8. Ans (A,B,D)



Let $x - 8 = 3\cos\theta$, $y - 6 = 3\sin\theta$

$$\text{Area} = \frac{1}{2}\begin{vmatrix} 0 & 8 & 1 \\ 6 & 0 & 1 \\ 3\cos\theta + 8 & 3\sin\theta + 6 & 1 \end{vmatrix} \quad \Delta = \frac{1}{2}$$

$[18\sin\theta + 36 + 24\cos\theta + 52]$

$$\Delta_{max} = \frac{1}{2}\left[52 + \sqrt{18^2 + 24^2}\right]$$

$= 41$

$\Delta_{max} = 41$, $\Delta_{min} = 11$

for $z_M + \bar{z}_M = 2\text{Re}(z_M) = \frac{104}{5}$

$$z_M = \left[\frac{52}{5}, \frac{39}{5}\right]$$

for $z_M + \bar{z}_M = 2\ \text{Im}(z_M) = \frac{78}{5}$

and slope of OC $< \frac{\pi}{4}$

9. Ans (A,C,D)

Put x = n and y = 1 then,

$$f(n+1) = \left(1+\frac{1}{n+1}\right) f(n) + \left(1+\frac{n}{2}\right)\frac{3}{2} + n^2 + 2n$$

Upon rearranging, one gets

$$\frac{f(n+1)}{n+2} - \frac{f(n)}{n+1} = n + \frac{3}{4}$$

Then we have

$$\frac{f(n)}{n+1} - \frac{f(n-1)}{n} = n - 1 + \frac{3}{4},$$
$$\frac{f(n-1)}{n} - \frac{f(n-2)}{n-1} = n - 2 + \frac{3}{4},$$
$$\vdots$$
$$\frac{f(2)}{3} - \frac{f(1)}{2} = 1 + \frac{3}{4}$$

Adding these equalities together, we get

$$\frac{f(n)}{n+1} - \frac{f(1)}{2} = 1 + 2 + ... + (n-1) + \frac{3}{4}(n-1)$$
$$= \frac{(n-1)n}{2} + \frac{3}{4}(n-1)$$

Thus,

$$f(n) = (n+1)\left[\frac{(n-1)n}{2} + \frac{3}{4}(n-1) + \frac{1}{2}\cdot\frac{3}{2}\right]$$
$$= \frac{n(n+1)(2n+1)}{4}$$

Now check the options

10. Ans (B,C,D)

$y' \sin x + y\cos x = x\sin x$

$\Rightarrow y' + y\cot x = x \Rightarrow y\sin x = \int x\sin x\, dx + c$

$\Rightarrow y\sin x = x\cos x + \sin x + c$

$\Rightarrow y = 1 - x\cot x \quad \left(\text{as } y\left(\frac{\pi}{2}\right) = 1\right)$

Now, $y'(x) = x\,\text{cosec}^2 x - \cot x$

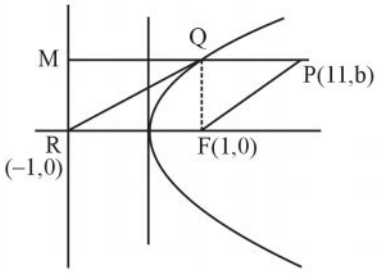
$$= \frac{x - \sin x\cos x}{\sin^2 x} = \frac{2x - \sin 2x}{2\sin^2 x} > 0 \ \forall\ x \in \left(0, \frac{\pi}{2}\right]$$

Hence y(x) is strictly increasing in $\left(0, \frac{\pi}{2}\right]$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (6)



$(FQ + QP)_{min} = (MQ + QP)_{min}$

M, Q, P must be collinear

$M(-1, 2t)$, $Q(t^2, 2t)$, $P(11,b)$

$$\Delta = \begin{vmatrix} -1 & 2t & 1 \\ t^2 & 2t & 1 \\ 1 & b & 1 \end{vmatrix} = 0$$

or $\boxed{b = 2t}$

If FQPR is a parallelogram then slope of FP = slope of QR

$$\frac{b-0}{10} = \frac{2t}{t^2+1} \Rightarrow \boxed{t = 3}$$

Area = 2 × 3 = 6

2. Ans (5)

$\det(A + xB) = \frac{1}{2}[(\text{Tr}(A + xB))^2 - \text{Tr}((A + xB)^2)]$

$= \frac{1}{2}[(\text{Tr}(A) + x\text{Tr}(B))^2 - \text{Tr}(A^2 + xAB + xBA + B^2x^2)]$

$= \frac{1}{2}[(\text{Tr}(A))^2 + 2\text{Tr}(A)\,\text{Tr}(B)x + (\text{Tr}(B))^2x^2 - \text{Tr}(A^2) - 2\text{Tr}(AB)x - \text{Tr}(B^2)x^2]$

$= \frac{1}{2}[(\text{Tr}(A))^2 - \text{Tr}(A^2)] + (\text{Tr}(A)\,\text{Tr}(B) - \text{Tr}(AB))x + \frac{1}{2}[(\text{Tr}(B))^2 - \text{Tr}(B^2)]x^2$

$= \det A + (\text{Tr}(A)\,\text{Tr}(B) - \text{Tr}(AB))x + (\det B)x^2,$

$272 = 22 + (28 - 33)x + 11x^2$

$\Rightarrow 11x^2 - 5x - 250 = 0$

$\Rightarrow x = -\frac{50}{11}$ or 5

3. Ans (6)

Let α be an integer satisfying the given equation, so that

$4\alpha^2 - (4\sqrt{3}+4)\alpha + \sqrt{3n} - 24 = 0$

$\Rightarrow 4\alpha^2 - 4\alpha - 24 = \sqrt{3}(4\alpha - n).$

Since $\sqrt{3}$ is irrational, it follows that

$4\alpha^2 - 4\alpha - 24 = 0$ and $4\alpha - n = 0$

Substituting $\alpha = \frac{n}{4}$ into the first equation, we have

$4\left(\frac{n}{4}\right)^2 - 4\left(\frac{n}{4}\right) - 24 = 0$

$\Rightarrow n^2 - 4n - 96 = 0$

$\Rightarrow (n - 12)(n + 8) = 0$

$\Rightarrow n = 12, -8$

Since n is a positive integer, we have n = 12

4. Ans (0)

$$I = \int_0^\infty \frac{1}{(1+x^{2024})(1+x^2)}\,dx \quad ....(1)$$

Put $x = \frac{1}{t}$

$$I = \int_\infty^0 \frac{x^{2024}.x^2}{(1+x^{2024})(1+x^2)}\left(-\frac{1}{x^2}\right)dx$$

$$I = \int_0^\infty \frac{x^{2024}}{(1+x^{2024})(1+x^2)}\,dx \quad ....(2)$$

$$2I = \int_0^\infty \frac{1}{(1+x^2)}\,dx = \frac{\pi}{2}$$

$I = \frac{\pi}{4}$ or $\left[\frac{\pi}{4}\right] = 0$

5. Ans (9)

There are $5 \cdot 6^3$ functions for which $f(1) \neq a$ and another $5 \cdot 6^3$ functions for which $f(2) \neq b$. There are $5^2 \cdot 6^2$ functions for which both $f(1) \neq a$ and $f(2) \neq b$.

So the total number of functions for which $f(1) \neq a$ or $f(2) \neq b$ or both is $5 \cdot 6^3 + 5 \cdot 6^3 - 5^2 \cdot 6^2 = 1260$

6. Ans (4)

Let A denote the event that the purchase accepts a lot. Now,

$P(A) = P(A|\text{lot has x defectives})\frac{3}{10} + P(A|\text{lot has 1 defective})\frac{7}{10}$

$$\frac{54}{100} = \frac{({}^xC_0)({}^{10-x}C_3)}{{}^{10}C_3}\left(\frac{3}{10}\right) + \frac{({}^1C_0)({}^9C_3)}{({}^{10}C_3)}\left(\frac{7}{10}\right)$$

$$\Rightarrow \frac{54}{100} \times {}^{10}C_3 = \frac{(10-x)(9-x)(8-x)}{6}\left(\frac{3}{10}\right) + \frac{9\times8\times7}{6}\times\frac{7}{10}$$

$$\Rightarrow \frac{54}{100} \times \frac{10\times9\times8}{6} \times 60 = (10 - x)(9 - x)$$

$(8 - x)(3) + 9 \times 8 \times 7 \times 7$

$\Rightarrow (10 - x)(9 - x)(8 - x) = 120$

$\Rightarrow x = 4$

7. Ans (4)

$T_n + 2 = 2\sqrt{2S_n}$

$\Rightarrow T_n^2 + 4T_n + 4 = 8S_n$ ....(1)

$\Rightarrow T_{n-1}^2 + 4T_{n-1} + 4 = 8S_{n-1}$ ....(2)

by (1) & (2)

$\Rightarrow T_n^2 - T_{n-1}^2 + 4(T_n - T_{n-1}) = 8(S_n - S_{n-1})$

$\Rightarrow T_n^2 - T_{n-1}^2 + 4(T_n - T_{n-1}) = 8T_n$

$\Rightarrow (T_n + T_{n-1})(T_n - T_{n-1}) = 4(T_n + T_{n-1})$

$\Rightarrow T_n - T_{n-1} = 4$

so sequence is A.P. with common difference 4

by equation (1) put $n = 1$, $S_1 = T_1$

$T_1 = 2$, $d = 4$

$\therefore T_{2024} = 2 + (2023)4 = 8094$

8. Ans (2)

Put $x = ty$

$$g(t) = \int_{-\frac{1}{t}}^{\frac{1}{t}} f(y)\,t\,dy = t\int_{-\frac{1}{t}}^{\frac{1}{t}} f(y)\,dy$$

$$\frac{g(t)}{t} = 2\int_{0}^{\frac{1}{t}} f(y)\,dy$$

$$= 2\int_{0}^{\frac{1}{t}} e^{-y}\,dy$$

$$= -2\left[e^{-y}\right]_0^{1/t} \Rightarrow -2\left[e^{-1/t} - 1\right]$$

$$\lim_{t\to 0^+} \frac{g(t)}{t} \Rightarrow 2$$